FRC 2024 CRESCENDO -- Robot Design Deep Dive

Each profile comes from the team's own public material: CAD (Onshape, GrabCAD, Fusion 360, STEP), technical binders, code, and Chief Delphi release and reveal threads. Performance data comes from The Blue Alliance.

The CRESCENDO design landscape

The game

CRESCENDO is a NOTE game. Robots collect NOTES (foam rings) from the floor or the SOURCE. Robots score NOTES in the AMP or shoot them into the SPEAKER. In the endgame, robots climb the CHAIN on the STAGE and can score a NOTE in the TRAP. Many designs used a speaker shooter with a fixed or variable angle, fed by an under-the-bumper intake. Many top teams did not build a TRAP mechanism.

How to use this document

The first section lists the profiled robots in the top 30 by Local EPA. The second section lists the other profiled robots. Each profile gives the archetype, the subsystems, vision, endgame, autonomous routines, the public CAD or binder, and lessons for students. A star marks a study pick. Each study pick has downloadable CAD or a full technical binder and one specific design lesson. "Unconfirmed" means the public material does not state the detail. It does not mean the robot does not have it.

About the EPA numbers

The Statbotics data API is offline. EPA is Local EPA (approx): it was computed from The Blue Alliance match data with the open-source Statbotics model (github.com/avgupta456/statbotics). The rank order follows Statbotics. The absolute values can differ from statbotics.io. Records are win-loss-tie totals from official events on The Blue Alliance.

Profiled robots at a glance (sorted by Local EPA)

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**3005 RoboChargers**	Surge	**54**	#4	59-10	4x NEO Vortex pivoting launcher with amp diverter; double-sided under-	CD thread
	4414 HighTide	TIDEPOD	**54**	#3	47-10	4-Kraken shooter on 4.5 in wheels; half-moon wheel sends notes to an a	CD thread
■	**1690 Orbit**	Doppler	**52**	#5	66-5-1	Low-profile robot; shooter pivot driven by machined pin-tooth pinions;	Onshape CAD
■	**2056 OP Robotics**	LOW-KEY	**52**	#6	67-3	Shooter on a low-backlash arm pivot; Limelight 3G 2D aiming; pneumatic	Tech binder
■	**604 Quixilver**	Domotron	**49**	#10	58-10-1	Shooter; chain-deployed intake with flexible polycarbonate rollers; el	Onshape CAD
■	**1778 Chill Out**	Zappy Feet	**49**	#11	43-22	Shooter with 2 bottom rollers and 1 top; amp bar on a virtual four-bar	Onshape CAD
■	**1706 Ratchet Rockers**	Riot	**48**	#16	44-4	Rack-and-pinion shooter hood; belt-driven elevator carrying the amp/tr	Onshape CAD
	2075 Enigma Robotics	Masquerade	**48**	#15	46-16-2	Note shooter with a shuttling shot mode; intake deploys from the start	CD thread
	4481 Team Rembrandts	CODA	**48**	#14	39-10	One shooter/amp 'Shamper'; two-sided under-bumper intake; dual climber	CD thread
	368 Team Kika Mana		**46**	#23	40-5	Speaker shooter with split wheels; hook climbers and a trap mechanism	CD thread
■	**2910 Jack in the Bot**	Typhoon	**46**	#22	50-16-1	+/- 360 deg turret shooter; two-entry feed loads notes at any turret a	Onshape CAD
	33 Killer Bees	Buzz XXIX	**45**	#30	49-17-2	4x Kraken X60 shooter with independent wheel sets, full-width complian	CD thread
■	**3467 Windham Windup**	Nocturne	**45**	#28	59-22-1	Arm and shooter; leaf-blower trap assist under the shooter; harmony cl	Onshape CAD
■	**1114 Simbotics**	Simbot Skyfall	**45**	#32	35-28	Version 2.0: rebuilt shooter that amps out the rear via a roller arm;	Onshape CAD
■	**3061 Huskie Robotics**	Artemis	**42**	#66	32-16-2	Pivoting flywheel shooter with kicker; intake with IR note sensing; sw	Onshape CAD
■	**581 Blazing Bulldogs**	Titan	**40**	#78	46-16	Pivoting split-flywheel shooter; angled amp/trap elevator; chain-drive	Onshape CAD
■	**4795 Eastbots**	Perseus	**39**	#93	47-13	22 lb pivoting shooter/amp tower on 2x NEO Vortex at 144:1 with gas sh	Onshape CAD
■	**6329 The Bucks' Wrath**	Dial	**39**	#95	53-19-1	High-pivot shooter that also scores amp and climbs; full-width under-b	Onshape CAD
■	**1787 The Flying Circuits**	Ronnie	**38**	#123	30-11	Pivoting flywheel 'accelerator' (1 Kraken per side); 'snake path' inta	Onshape CAD
■	**5951 Makers Assemble**		**38**	#113	43-21-2	Fixed-speed shooter; under-bumper intake; pivot arm; no climber; swerv	Drive folder / doc
■	**5419 Berkelium**	Radium the Roomba	**37**	#132	26-14-1	Elevator + arm shooter; 4-side ground pickup via street sweepers; Amp	Onshape CAD
■	**131 CHAOS**	Double Time	**36**	#144	45-25-1	Launcher scores Speaker, Amp and Trap; under-bumper intake; lift with	Onshape CAD
■	**3357 COMETS**	Vaisala	**36**	#150	47-15	Roller-pinion sector gears; silicone-over-tube rollers; 14 SPARK MAX;	GrabCAD
	4043 NerdHerd	Notezart	**36**	#145	46-16	RI3D-style arm with flywheels and kick-up feed wheels; intake in a pro	Onshape CAD
	5406 Celt-X		**36**	#159	41-21	Flywheel shooter set by arm angle; roller intake; amp scoring; climber	Onshape CAD
■	**6036 Peninsula Robotics**	Snoopy	**36**	#162	43-19-1	Turreted speaker shooter on an X-contact bearing with auto-aim	Onshape CAD
■	**111 WildStang**		**35**	#180	22-23	Tilting shooter with note compression; lift on the amp note path; LL3	Onshape CAD
■	**2877 LigerBots**		**35**	#192	35-27-1	Shooter pivots on an angled cascade elevator; under-bumper intake; spl	Onshape CAD

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**9312 NERD Spark**	BATMAN	**35**	#187	33-23	Intake-shooter tilted by a printed CF 4-bar; 2 DOF arm with wrist for	Onshape CAD
■	**8033 Highlander Robotics**	Banshee	**34**	#208	33-13	Pivoting shooter on a Maxspline shaft, 2 independent wheel sets; UTB i	Onshape CAD
■	**341 Miss Daisy**	Miss Daisy XXII	**33**	#247	38-22-1	Full-width over-bumper intake feeding an under-bumper intake; belts on	Onshape CAD
■	**2930 Sonic Squirrels**	Maestro	**33**	#238	29-26-1	Continuous belt-driven elevator; PhotonVision note detection and local	Onshape CAD
■	**930 Mukwonago BEARs**		**32**	#290	35-16-1	Turreted shooter driven through a strain-wave gearbox; 3D printed gear	Onshape CAD
■	**2102 Team Paradox**		**32**	#262	19-10	Combined intake/shooter with two roller pairs (shoots from either side	Onshape CAD
	7312 T3		**32**	#293	37-22	Fixed-angle two-sided shooter, rotary amp bar, OTB rotary intake, Grey	Onshape CAD
	1710 The Ravonics Revolution	ADONIS	**31**	#316	29-13	Wrist and shooter; intake; rope-rigged elevator; Stage chain climb; sw	Fusion 360 CAD
■	**1155 SciBorgs**	Taiwanese Fried Chicken	**30**	#334	18-9	Pivoting flywheel shooter that also scores Amp; shooting on the move f	Onshape CAD
	3136 O.R.C.A.	JT	**30**	#367	38-23-1	Arm-mounted intake shoots out the back with two flywheels; AprilTag di	Onshape CAD
■	**3729 The Raiders**	Mr. Mimlitz	**30**	#328	15-13-1	Speaker and amp; hybrid under-bumper intake with slapdown and printed	Onshape CAD
	5431 Titan Robotics	Orpheus	**30**	#341	22-23	V2 rack-and-pinion shooter angler (about 35-55 deg); pivot intake scor	Fusion 360 CAD
■	**1414 IHOT**	Encore	**29**	#420	41-21	Orbit-inspired layout: differential speed shooter; combined intake/ind	Onshape CAD
■	**6647 PrepaTec - VOLTEC**		**29**	#415	34-21-1	Kraken shooter and intake; elevator removed mid-season; swerve	Onshape CAD
■	**2491 NoMythic**	Eurydice	**27**	#475	17-8	Bottom-pivot two-roller shooter that also scores amp; under-bumper int	Onshape CAD
■	**7525 Pioneers**		**27**	#523	22-13-1	Static shooter; beater-bar intake; roller amp mechanism added for Worl	Onshape CAD
	1038 Lakota Robotics	Vio-Linda	**25**	#663	14-16	Flywheel launchers and a 'TRAG' amp/trap scorer that feeds them; under	Onshape CAD
	3749 Team Optix		**25**	#622	17-9	Single-jointed arm carrying a four-bar intake that also shoots; two Li	Onshape CAD
■	**6838 X-SHARC**	Blackout	**25**	#660	15-11	Shooter on a pivoting arm; touch-and-grab ground intake; amp shot back	Onshape CAD
	2423 The KwarQs	ConDUCKtor	**24**	#731	20-23	One pivot for shooter and climber; over-the-bumper intake; swerve	Onshape CAD
■	**5987 Galaxia in memory of David Zohar**	Andromeda	**24**	#693	30-26	Polycarbonate spring-hook Stage climber with a locking cable winch; fl	Onshape CAD
■	**3737 Roto Raptors**	Synthwave	**22**	#850	25-18-2	Through-robot note path; under-bumper intake; sprung single-hook climb	Drive folder / doc
■	**6925 W.A.Robotics**	VENDETTA	**22**	#878	26-21	Fixed-angle 2-Kraken shooter; over-the-bumper intake with handoff; two	Onshape CAD
■	**4276 Surf City Vikings**		**19**	#1150	15-9-1	Intake/shooter on a NEO-driven 4-bar, rebuilt mid-season; floor intake	Onshape CAD
■	**663 Power Sharks**	Hammerhead	**17**	#1418	18-23	Ri3D-based wide intake-shooter; sliding 8020 extending climbers on #35	Onshape CAD
	1072 Harker Robotics		**16**	#1561	11-19	Pivoting top-down shooter on a two-stage cascade elevator; over-the-bu	Onshape CAD
■	**1746 Team OTTO**		**16**	#1591	30-17	Falcon shooter fed by a primer; intake rebuilt on a Falcon; CTRE swerv	Onshape CAD
	9047 SESI SC TECHMAKER	Seu Wilson	**15**	#1620	7-3-1	Fixed under-bumper intake-shooter; elevator that climbs and carries an	Onshape CAD
■	**3596 Rocket Robotics**	Kerchunk	**11**	#2329	3-7	Kitbot launcher; under-bumper ground intake; AndyMark Climber in a Box	Onshape CAD

■ = study pick: downloadable CAD or a full technical binder, and one specific design lesson. EPA is Local EPA (approx), computed from The Blue Alliance match data. Records are win-loss-tie totals from official events.

Top-30 robots by Local EPA

■ **Study pick** | **3005 RoboChargers** Surge | USA-TX | **Local EPA 54 (#4)** | 59-10

4x NEO Vortex pivoting launcher with amp diverter; double-sided under-bumper intake; telescoping climber; swerve

SHOOTER (SPEAKER) Pivoting launcher with four NEO Vortex motors, chosen for mechanical simplicity: there is no mechanical linkage between the four corners of the launcher. A speed difference between the two sides puts spin on the note. Surgical tubing counterbalances the launcher.	**INTAKE** Under-the-bumper, double-sided intake. The intake was upgraded several times during the season and moved between the competition robot, the practice robot and a spare intake.
FEEDER / AMP / TRAP A diverter on the launcher allows both amp and speaker shots. A team member mentions a feeder motor that was never removed during the season.	**DRIVETRAIN** Swerve with NEO Vortex motors. The team rotates used swerve modules into the spares pool so spares are 'known good'. A team member credits SPARK Flex controllers with fewer module and motor swaps this season.
ENDGAME Single telescoping climber. One part on the robot keeps notes from getting stuck on the climber.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** CAD release (CD)

LESSONS FOR STUDENTS

- Block CAD released next to the finished robot: open both Onshape documents to see how the layout model drives each subsystem. The team states that links from the robot to the block CAD cannot be opened by right-click.
- Four independent launcher motors remove the mechanical linkage between the launcher corners. The team states the same spin could be made without four motors, so the choice was for mechanical simplicity.
- SRPP as a polycarbonate replacement: the team lists lower density, higher tensile stiffness, Loctite resistance and laser cutting as the reasons. The team also states it gives little to no accuracy benefit over polycarbonate.
- A spares practice: used swerve modules move into the spares pool so that spares are known to work.
- The block CAD method is documented in a FIRST Mentor Conference talk by a 3005 member, with video and slides.

CD thread: CAD release (CD) · CAD release (CD) · Top-level robot CAD (Onshape) · Block CAD (Onshape) · Reveal 'Surge' (CD) · Reveal video · Block CAD talk (CD) · Block CAD talk video · Block CAD talk slides (Google Slides)

Profile	4414 HighTide TIDEPOD USA-CA	Local EPA 54 (#3) 47-10

4-Kraken shooter on 4.5 in wheels; half-moon wheel sends notes to an amp elevator

SHOOTER (SPEAKER) Two-sided wheel shooter. Wheels are 4.5 in lexan tube with 3D printed end caps, wrapped in cat's tongue. 1.5 in compression, 5 in center-to-center, and 1:1.25 upduction on 4 Krakens, 2 per side, with each side linked. Double-sided HTD 5 mm belts drive the wheels. There is no side-to-side compression; most shots use about a 5% speed difference between sides.	**INTAKE** Unconfirmed. A team post on 2024-05-03 describes a mechanism based on the '254-2022 cube intake' but does not name the subsystem. That mechanism uses laser-cut 7075 plates, roller bearings and delrin, packaged around a 2x1 tube.
FEEDER / AMP / TRAP A belted half-moon wheel in the shooter has three states. It hard-stops notes in the shooter, flips up out of the way to shoot, and spins in reverse to send the note over the top of the shooter wheel into the amp elevator. It is belted 1:1 off a Kraken, so the motor's internal absolute encoder handles position control. The team calls it 'hacked in'; the belt slipped once, in the last match of the season, and hurt shooting.	**DRIVETRAIN** Unconfirmed
ENDGAME Unconfirmed	**AUTONOMOUS** The team won the Autonomous Award at the Ventura County Regional.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Robot reveal (CD)

LESSONS FOR STUDENTS

- A shooter specified in one post, with wheel diameter, material, compression, center-to-center, ratio and motor count.
- Retest a borrowed trick on the new geometry: reversing the shooter wheels lifted notes on the team's turret robot but not with 4.5 in wheels, so the team added a half-moon wheel.
- A mechanism belted 1:1 to its motor can use the motor's internal encoder for position control, as the half-moon does. A slipped belt then costs accuracy, as it did in the team's last match.
- Tension belts by reprinting: idlers are toothless pulleys on shoulder bolts, and the team reprints an idler 2-3 times after each spacing or ratio change.
- Match parts to build speed: the team kept a traditional bearing stack instead of a large X-contact bearing because it builds the robot in under a week after design lock.

CD thread: Robot reveal (CD) · Robot reveal (CD) · Reveal video · Robot teaser (CD)

■ Study pick

1690 Orbit Doppler

Israel

Local EPA 52 (#5)

66-5-1

Low-profile robot; shooter pivot driven by machined pin-tooth pinions; amp and trap mechanisms; swerve

SHOOTER (SPEAKER)
Shooter on a pivot. The pivot pinions are machined from two plates with steel pins between them that act as teeth. The shooter is removable: clamshell plates with interlocking 'Lego'-style edges clamp it to custom blocks on the main shaft; the team says this is more rigid than screws alone. Two mechanical stops below the shooter edges hold it in travel mode, and a Hall effect sensor detects the arm at the bottom of its travel. A team member wrote that other teams' spin shooters worked better than Doppler's.

INTAKE
A team member wrote that other teams' larger-diameter intakes worked better than Doppler's.

FEEDER / AMP / TRAP
The robot has an amp mechanism and a trap mechanism. In the trap mechanism a servo pulls a thin 3D-printed lever to release the arm. A constant force spring pulls the arm forward, and a torsion spring in its 3D-printed pivot rotates it. A team member wrote that downward-shooting amp mechanisms, and trap mechanisms that get closer to the trap, worked better than Doppler's.

DRIVETRAIN
Swerve. The team uses an MA3 absolute encoder on its swerve modules.

ENDGAME
The robot has a trap mechanism with a servo release and spring-driven arm. The robot slows its drive in the climb state.

AUTONOMOUS
The team ran a 7 note auto. Paths come from the team's own Pathfinder tool. It writes a CSV that the robot parses before each match, with velocity and acceleration limits. Note detection lets the robot skip a missing midline note: in the auto run through most of the Championship playoffs, the robot looks for the first midline note; if it is absent, it jumps ahead and drives to the third. Won the Autonomous Award at ISR District Event #1.

VISION
Three cameras. A Limelight at the back (shooter side) and an Arducam running PhotonVision at the front read AprilTags. A second front camera detects notes. The June 2024 software sessions covered pose tracking and shooting on the move. Note positions come from the camera's Tx and Ty with trigonometry. The team scales its own odometry by a carpet factor measured in practice matches: drive the field end to end and compare.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- Pin-tooth pinions drive the pivot; the team says they are less complex to make than they look.
- Build a fallback into an auto path: when the camera does not see the first midline note, Doppler's auto jumps to a later path point and goes for the third note.
- Calibrate odometry to the carpet with a factor measured in a practice match.
- Integrate CAD often: the team defines subsystem volumes first, gives each subsystem a CAD lead, and integrates about 3 times a day.

Onshape CAD: Onshape CAD · Onshape CAD · Top-level x_t files (Google Drive) · CAD release thread (CD) · Behind the Bumpers video · Reveal thread (CD) · Software session slides (Google Slides)

■ Study pick

2056 OP Robotics LOW-KEY

Canada

Local EPA 52 (#6)

67-3

Shooter on a low-backlash arm pivot; Limelight 3G 2D aiming; pneumatic climber; swerve

SHOOTER (SPEAKER) Upper and lower flywheels; the arm sets the shot angle. After the first district event the team tipped the flywheels downward for a lower release angle, so the robot could score from the wing line; the arm could not go low enough for that shot. The feed position did not change, so the note handoff was not affected.	**INTAKE** Rollers of 1 in OD, 3/8 in ID 40A polyurethane stretched onto live 1/2 in rounded hex shafts. Timing pulleys in the middle of the rollers let the timing belts act as conveyor belts. Side kicker rollers are 2 in OD stealth wheels. One Kraken X60 drives the intake through a 2:1 reduction for a linear speed of 25 ft/s.
FEEDER / AMP / TRAP Shooter feed rollers are blue 1 in OD sushi rollers on a 1/2 in hex shaft. The arm drives to an amp scoring position with CTRE Motion Magic and tracks the speaker with a PD loop with no gravity term. The arm gearbox is custom, with every gear bonded to its shaft with Loctite 609.	**DRIVETRAIN** Swerve. In a team post of 2024-10-25 on module wiring, CAN runs from the first motor to the encoder, to the second motor, then to the next CAN device. CAN and power wires pass through a cut-out in the top module cover and are soldered to the CANcoder board.
ENDGAME The climber uses pneumatic quick exhaust valves, which the team buys from Automation Direct.	**AUTONOMOUS** In auto, chassis yaw, tx and distance to the tag correct the odometry estimate. The code uses the WPILib timed framework, not command-based. TBA lists the Autonomous Award at McMaster, Newmarket and the Ontario Science Division.
VISION Limelight 3G on LimelightOS with no coprocessor. It tracks the center speaker AprilTag in 2D: tx turns the chassis to aim and ty gives distance, which sets the arm angle. No MegaTag or 3D pose estimation. The team states that 2D tracking runs at a higher FPS and needs no filtering or latency compensation.	**PUBLIC CAD / DESIGN** Technical binder (PDF)

LESSONS FOR STUDENTS

- Backlash control without shim tape: Loctite 609 fills the gap in each gear bore, and the final stage uses a broached sprocket and a 1 in long machined gear hub instead of a 1/2 in deep gear bore.
- Change the cheap part first: tipping the flywheels downward lowered the release angle for wing-line shots without a new arm range or a new note handoff.
- Simple vision can be enough: 2D tracking of one AprilTag aimed the chassis and set the arm angle, with no pose filtering or latency compensation.
- Match the controller to the move: Motion Magic for large arm moves and a PD loop for tracking a moving setpoint.
- The team states why it builds custom gearboxes instead of planetaries: custom packaging and chain tensioning, in-house manufacturing, and a way to remove backlash in a long arm.

Tech binder: Technical binder (PDF) · Technical binder release thread (CD) · Technical binder (PDF) · CAD drawing package (PDF) · Robot teaser video

■ Study pick

604 Quixilver Domotron

USA-CA

Local EPA 49 (#10)

58-10-1

Shooter; chain-deployed intake with flexible polycarbonate rollers; elevator; climber

SHOOTER (SPEAKER) The shooter hex shafts were cut shorter on the robot than the CAD shows.	**INTAKE** Deployable intake with polycarbonate rollers that flex in an impact; a roller could flex more than half its diameter in a bad hit. Short shaft segments replace a full-width hex shaft, which would bend permanently. The left and right sides are tied together through chain and a cross shaft inside the frame perimeter, so no rigid cross bars are used. The team chose 3/8 in bearings for weight and fit in the tube plugs, and used 3/8 in OD spacers in place of hex. The team reports two failure points it did not solve in season: the bolts holding the axles to the side plates sheared, and the deploy chain snapped.
FEEDER / AMP / TRAP The robot has an elevator. Two hex shafts on the elevator came from an earlier climber iteration.	**DRIVETRAIN** The bellypan holes were hand-drilled, not CNC-cut, and the CAD omits most of them. Electronics are suppressed in the CAD to reduce lag; the team planned the layout in advance.
ENDGAME The robot has a climber. The team reused the top elevator hex shaft to tension the climber with surgical tubing. The tubing is not in the CAD.	**AUTONOMOUS** The team won the Autonomous Award in the Milstein Division.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Design the intake to flex instead of resisting impact: polycarbonate rollers on short shaft segments can deflect more than half their diameter without bending a shaft.
- Tie the two sides of a wide intake together through a cross shaft and chain inside the frame perimeter, so the rollers need no rigid cross bars.
- Record the failure modes you did not fix. The team lists sheared axle bolts and a snapped deploy chain as open problems for a similar future intake.
- Compare the CAD with the built robot. The team states where they differ: hand-drilled bellypan holes, shorter shooter shafts, and climber tensioning by surgical tubing.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video

■ Study pick

1778 Chill Out Zappy Feet

USA-WA

Local EPA 49 (#11)

43-22

Shooter with 2 bottom rollers and 1 top; amp bar on a virtual four-bar with no extra motor

SHOOTER (SPEAKER) Shooter with two rollers on the bottom and one on the top. Shooter compression is 1/2 in, with no horizontal compression. The team added the second bottom roller because a one-roller prototype shot above the amp bar; it also made intaking faster.	**INTAKE** Intake compression is 0.2 in. The two bottom shooter rollers help bend the note into the shooter when intaking.
FEEDER / AMP / TRAP Amp scoring uses an amp bar on a virtual four-bar, which removed the need for an extra motor on the shooter subsystem. The design lead added it after the team tested an idea from RI3D. The reason was concern that notes would fall out of the amp, possibly into the bellypan.	**DRIVETRAIN** Unconfirmed
ENDGAME Unconfirmed	**AUTONOMOUS** Autonomous Award at PNW District Bonney Lake.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A linkage can replace an actuator: the amp bar on a virtual four-bar removed an extra motor from the shooter subsystem.
- A prototype result can set roller layout: a one-roller shooter shot above the amp bar, so the team added a second bottom roller.
- Write down compression for the shooter and the intake.
- Process before CAD: the team set priorities and strategy first, sketched several archetypes, and narrowed to one; the final robot stayed close to the first KrayonCAD.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub)

■ Study pick

1706 Ratchet Rockers Riot

USA-MO

Local EPA 48 (#16)

44-4

Rack-and-pinion shooter hood; belt-driven elevator carrying the amp/trap mechanism; Thrifty climber kit; swerve

SHOOTER (SPEAKER) The shooter hood moves on a rack and pinion. It ran on a P term only; the team states it should add an arm feedforward (kG/kS) because the hood has a swinging CG. The rack skipped teeth until the team increased the DP mid-season and changed to a steel shaft (drop-in, 3x stiffness).	**INTAKE** Unconfirmed
FEEDER / AMP / TRAP An elevator carries the amp and trap mechanism. The two sides are belt driven with no mechanical link between them, so the team tuned belt turnbuckles to stop racking and skipped teeth. Control uses the WPILib ElevatorFeedForward and TrapezoidProfile classes. The team changed the elevator to NEO 550s after a failure at Roboteer Rumble (NEO encoder wires and a melted pulley).	**DRIVETRAIN** Swerve. A reply noted plate construction in the bracing and cross members; the team said these parts were machined on its Tormach PNC 770. The team reports zero structural failures anywhere on the robot all season.
ENDGAME The climber is the Thrifty climber kit. For the trap, the driver climbs and then holds Y to drive the elevator to max height, where the motor stalls at full extension. The trap code is not in the public repository.	**AUTONOMOUS** Autonomous Award at the St. Louis Regional.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Unlinked elevator sides need tuning: the team adjusted belt turnbuckles so the elevator would not rack or skip teeth, and states it would design this differently.
- A rack-and-pinion hood can skip teeth; the team fixed it mid-season by increasing the DP and changing to a steel shaft.
- Bushings can replace bearings at moderate speed: the team believes most ratios are about 3:1 off the motors, 'roughly 1500-2000' (no unit given), with axial float, no tight press fits and oil worked in after assembly. The team replaced no bushings all season.
- The elevator runs WPILib ElevatorFeedForward and TrapezoidProfile instead of Smart Motion; the team states it would use MotionMagic on a Kraken.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Code (GitHub) · Reveal 'Riot' (CD) · Reveal video

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Profile

2075 Enigma Robotics Masquerade

USA-MI

Local EPA 48 (#15)

46-16-2

Note shooter with a shuttling shot mode; intake deploys from the start configuration; Trap added mid-season

SHOOTER (SPEAKER) Note shooter with a shuttling shot mode added before Kentwood.	**INTAKE** The reveal photo caption shows the intake releasing from its start-of-match configuration. The team reports improved intake reliability before Kentwood.
FEEDER / AMP / TRAP Trap scoring was added before Kentwood. The trap and amp hardware is not described.	**DRIVETRAIN** Unconfirmed
ENDGAME The team redesigned the whole climb system before Kentwood and added Trap scoring. The team reports 9 of 12 Traps in MSC qualifications and 4 of 4 in playoffs.	**AUTONOMOUS** Before MSC the team added new center autos and tuned automatic note pickup and auto localization.
VISION Limelight 3G (LL3G), adopted before FIM Kentwood. Note detection drives automatic pickup. Localization during autonomous was tuned before MSC.	**PUBLIC CAD / DESIGN** Reveal thread with season updates (CD)

LESSONS FOR STUDENTS

- Mid-season changes can be large: between week 1 and week 5 the team switched to LL3G vision, redesigned the climb and added a Trap.
- Track mechanism success rates, as the team reports for Traps at MSC.
- Note detection for automatic pickup was added in season, then tuned before MSC.

CD thread: Reveal thread with season updates (CD) · Reveal thread with season updates (CD) · Reveal video · Video linked in the reveal thread

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Profile	4481 Team Rembrandts CODA Netherlands	Local EPA 48 (#14) 39-10

One shooter/amp 'Shamper'; two-sided under-bumper intake; dual climbers; REV MAXSwerve; 7-note auto

SHOOTER (SPEAKER) Combined amp and speaker scoring system that the team calls 'The Shamper': a shooter with an amping mechanism. Rollers run on individual RPM setpoints. Auto aim uses lookup tables; the team chose them because they are easy to adjust live. For moving shots, a team member writes he is '99% certain' the code compensated for robot velocity by using a position further 'into the future'.	**INTAKE** Two-sided under-the-bumper intake.
FEEDER / AMP / TRAP Amp scoring uses the same 'Shamper' system as the speaker shot.	**DRIVETRAIN** REV MAXSwerve modules. The team updated 'all our SPARK Flex controllers' in April 2024; the post does not say which subsystems use them.
ENDGAME Dual climbers.	**AUTONOMOUS** 7-note auto, tested in April 2024 before the Championship. Changes to roller setpoints, Shamper wheels and grip tape did not improve long shots. The team then moved the centerline shooting point toward the center of the stage. In 8 timed test runs, 2 scored 7 notes and 6 scored 6. The team states it is not 100% consistent.
VISION PhotonVision. The default WPILib SwerveDrivePoseEstimator fuses vision tag readings with odometry, using vision standard deviations. The OP thinks it uses odometry until a tag is detected and the pose difference is not too large, and adds that he could be wrong.	**PUBLIC CAD / DESIGN** Open Alliance build thread (CD)

LESSONS FOR STUDENTS

- Compare 'The Shamper', one system that scores in both the amp and the speaker, with robots that use a separate amp mechanism.
- The 7-note auto post shows a test method: 8 timed runs with the same notes, each run listed with notes scored and a reason where a run fell short. Students can copy the table format.
- When shooter changes did not fix long auto shots, the team moved the shot closer and accepted the time cost. The post states the trade.
- The team logs sensor data, pose, camera readings, subsystem states and controller inputs, and plots current draw per motor per state in AdvantageScope to prevent brownouts.
- Auto aim uses lookup tables, and a team member believes moving shots used a projected future position. He states he prefers a tunable method over one that works better on paper.

CD thread: Open Alliance build thread (CD) · Open Alliance build thread (CD) · Reveal 'CODA' (CD) · Reveal video · State machines post (CD) · Code (GitHub org)

Profile	368 Team Kika Mana USA-HI	Local EPA 46 (#23) 40-5

Speaker shooter with split wheels; hook climbers and a trap mechanism

SHOOTER (SPEAKER) Speaker shooter with split wheels. The split wheels put a little spin on the note to stabilize the shot, an idea taken from an Open Alliance build. After Los Angeles the team added a second shooter motor and raised wheel speed for a more accurate wing-line shot, mostly for auto.	**INTAKE** After Los Angeles the team rebuilt the front end and made new bumpers. The OP states the intake was no match for the stage.
FEEDER / AMP / TRAP Unconfirmed	**DRIVETRAIN** Unconfirmed
ENDGAME Hook climbers on the stage chain and a trap mechanism. After Los Angeles the team pre-deployed the hook climbers and raised speed by 20%, aiming for a trap under 3 seconds.	**AUTONOMOUS** The team added auto modes after Los Angeles to fit with alliance partners. Autonomous Award at the Hawaii Regional.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Reveal thread with spec sheet (CD)

LESSONS FOR STUDENTS

- Iterate between events with a short change list: the team named four changes after Los Angeles (climber pre-deploy, a second shooter motor, more auto modes, a more robust front end).
- Split shooter wheels add spin to stabilize the note; the OP credits the idea to an Open Alliance build.
- Design the front end for contact with field structures: the OP states the intake 'was no match for the stage', and the team made the front end more robust.
- No CAD or code is public; the spec sheet PDF and the reveal video are the study material.

CD thread: Reveal thread with spec sheet (CD) · Reveal thread with spec sheet (CD) · Reveal video

■ Study pick

2910 Jack in the Bot Typhoon

USA-WA

Local EPA 46 (#22)

50-16-1

+/- 360 deg turret shooter; two-entry feed loads notes at any turret angle; about 16.5 in tall; no Trap

SHOOTER (SPEAKER) Turret shooter. The turret rotates +/- 360 degrees and loads notes from any angle in that range. The robot is about 16.5 in tall with the shooter down.	**INTAKE** The intake is modeled after the Spectrum intake and picks up notes from many angles.
FEEDER / AMP / TRAP Load-from-any-angle feed. The team compares it to the 254 spindexer with two entry points: the note enters from the 180 degree arc facing the intake. Feeding takes slightly longer with the turret at 90 or 270 degrees, because the teacup rollers contact the note as it enters the indexing zone. The team did not design in a Trap function. At reveal, a team member wrote that the amp plan still had issues to resolve.	**DRIVETRAIN** Unconfirmed
ENDGAME Changed after the reveal. The first plan was a spring-powered polycarbonate hook released by a solenoid; the pivot would then pull down to climb. In testing, the pivot motors in brake mode could not hold the robot up long enough after the match. The hook sat further from the center of gravity than planned, so the robot hung at a steep angle. The turret motor brake could not hold the turret, so the base could rotate and might touch the ground (team post 2024-04-13). The released robot uses a climber gearbox that put a high load on its motor. After some successful climbs the motor stalled at low voltage and current. The team believed new motor protection firmware caused this. After a firmware rollback the motor drew about 180 amps, and one burned out at DCMP even with an e-stop after each climb. Without a root cause, the team recommends other teams explore alternate designs.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A spindexer-style feed with two entry points takes the note from the 180 degree arc facing the intake.
- Measure the cost of the worst-case angle: the team states feeding is slightly slower at 90 and 270 degrees.
- Check holding torque before you rely on motor brake mode; in tests, brake mode could not hold the robot or the turret.
- Size climber motors with margin: this climber motor burned out at DCMP.
- Publish unresolved failures with the CAD: the team released the climber problem without a root cause and advised other teams to try other designs.

Onshape CAD: Onshape CAD · CAD, code and tech binder release (CD) · Onshape CAD · Code (GitHub) · Tech binder (Google Drive) · Robot reveal (CD) · Reveal video

Profile	33 Killer Bees Buzz XXIX USA-MI	Local EPA 45 (#30) 49-17-2

4x Kraken X60 shooter with independent wheel sets, full-width compliant intake, side chain climb

SHOOTER (SPEAKER) A new shooter ran at Livonia, the Michigan State Championship and the Championship. It uses 4x Kraken X60, with each set of wheels driven independently. A polycarbonate, Teflon-skinned tab-and-slot box compresses the note side to side to 12 in. The team says this increased consistency when shooting from back at the wing line.	**INTAKE** Full-width compliant front roller. It replaced vertical flappers that worked in the shop; after week 1, the team reports that new notes on new carpet slid 'like hockey pucks on ice'.
FEEDER / AMP / TRAP The team had planned to score in the trap and dropped it right before the Kettering event.	**DRIVETRAIN** Unconfirmed
ENDGAME Side chain climb, added for Livonia. A mast similar to the team's 2020 robot carries a spring-loaded hook that pops out and grabs a single chain link. The team ties the climb to ranking higher after it dropped the trap.	**AUTONOMOUS** In some Championship matches the auto path ended behind the truss and the robot still made its shots.
VISION Limelight cameras. A fourth Limelight, mounted in the center of the robot, was added for the Michigan State Championship for TargetPose. The team reports that targeting and localization software changes between MSC and the Championship increased shot accuracy in auto and teleop.	**PUBLIC CAD / DESIGN** Robot reveal and update thread (CD)

LESSONS FOR STUDENTS

- A shooter box that compresses the note side to side, with independently driven wheel sets, is the team's stated change for consistent shots from the wing line.
- Test intakes on worn and new field conditions: flappers that worked in the shop failed with new notes on new carpet.
- Scope can change mid-season: the team dropped the trap before its first event and added a fast side climb to rank higher.
- The team reports that a fourth Limelight and late software work on localization raised shot accuracy between MSC and the Championship.

CD thread: Robot reveal and update thread (CD) · Robot reveal and update thread (CD)

■ Study pick

3467 Windham Windup Nocturne

USA-NH

Local EPA 45 (#28)

59-22-1

Arm and shooter; leaf-blower trap assist under the shooter; harmony climb hooks

SHOOTER (SPEAKER) The trap post names an arm with uprights and a shooter above the blower. Whether the shooter rides on the arm is unconfirmed. The team states its cleanup play often relied on the subwoofer shot.	**INTAKE** The robot browned out at the end of two Championship playoff matches. The team states that front-court play ran the shooter and intake motors much more than cycling did.
FEEDER / AMP / TRAP Trap: a converted Milwaukee M12 Compact Spot Blower blows the trap door open so the robot can shoot a note into the trap. The team took the concept from 4028's reveal. It replaced the blower's non-FIRST-legal 775 motor with a 775 Pro and press-fit the original output shaft onto it. At the Championship the blower trap scored in a practice match but not in a qualification match. The team states it worked no better at the NE District Championship.	**DRIVETRAIN** In December 2023 the team planned to run SDS MK4i swerve modules again, with swerve code derived from Team 364's BaseFalconSwerve. The Championship recap names a 'Front Left Steer' CAN device in the RIO logs.
ENDGAME The robot used harmony climb hooks. At the Championship it harmony climbed three times: QM71 and both the original and replay of QM110, which earned a ranking point. On kickoff day the team listed the trap as TBD on its priority list. The blower was on the robot from Greater Boston onward.	**AUTONOMOUS** Rush autos to the centerline notes. In Championship playoff matches the team ran a 'Poop n' Scoot' auto from the Amp side that reached the centerline note in about 2.33 seconds. The team chose the next note from its prediction of the opponents' autos. The team states these autos were fast but not always accurate. Autonomous Award at North Shore and at the NE District Championship Ganson Division.
VISION The team tested the blower trap shot with auto align. Its next step was to improve AprilTag pose estimation accuracy for shot repeatability. In December 2023 the team planned AprilTag work with one or more Limelights.	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape)

LESSONS FOR STUDENTS

- A trap mechanism from a power tool: a converted Milwaukee M12 blower.
- Model purchased parts at true size: the team states the CAD blower matches the length and diameter of the blower with its twist-on extension.
- Plan for wear: the team states Nocturne struggled to stay on the field from the end of the NE Championship. By the off-season it had played a little over 110 matches. It then built 'Black Cherry', a clone of 1678's robot, to practice an elevator and an over-bumper intake.
- Leave slack on every CAN wire: in a Championship playoff match a wire with a snapped Molex SL crimp clip, routed under the arm gearbox with little slack, made the robot stop moving at intervals. The team found it from a CAN Stale Error in the RIO logs.

Onshape CAD: Full robot CAD (Onshape) · Full robot CAD (Onshape) · Trap blower thread (CD) · Build blog with Championship recap (CD) · Off-season 'Black Cherry' write-up (CD)

Other profiled robots

Each has public CAD, a binder, code or a detailed release thread. Sorted by Local EPA. ■ marks the study picks.

■ Study pick **1114 Simbotics** Simbot Skyfall
Canada

Local EPA 45 (#32)
35-28

Version 2.0: rebuilt shooter that amps out the rear via a roller arm; chain climb with a latch

SHOOTER (SPEAKER) Version 1.0 used a sliding shooter. The team states the slide slowed cycle times and the shooter lacked accuracy. Before McMaster the team redesigned and rebuilt the shooter and the climber. Barlow helped make new sheet metal parts for the redesigned robot.	**INTAKE** The original intake system used the same outer-roller-faster gearing.
FEEDER / AMP / TRAP 2.0 scores the amp out of the rear of the shooter through an arm. The note turns inside the arm, so the outer rollers need a higher surface speed than the inner ones: 3.11:1 between top and bottom roller. A light sensor in the arm lets the robot partly eject the note, align it with the amp opening, then eject it fully. Amp geometry came from sketches, not a prototype.	**DRIVETRAIN** Unconfirmed
ENDGAME Stage chain climb. The team states the 1.0 climber was too complex to give consistent results, and 2.0 replaced it. The 2.0 arm did not stay up long enough for the climb to count, so team members held an internal contest to design a latch; the winning design went on the robot. The arm must swing onto the chain at a tuned speed for the latch to engage.	**AUTONOMOUS** The team cites a Milstein playoff match with 4 notes scored in auto (30 in total).
VISION After the Ontario Provincial Championship the team added vision tracking that detects game pieces and corrects the robot's path to them in autonomous and teleop.	**PUBLIC CAD / DESIGN** Simbot Skyfall 2.0 CAD (Onshape)

LESSONS FOR STUDENTS

- When a note turns through a roller path, run the outer rollers faster to stop jams.
- A light sensor can stage a note for the amp: partly eject, align with the opening, then eject fully.
- A climb that does not hold can be fixed with a latch; this latch needs the arm to swing onto the chain at a tuned speed.
- With no time to prototype, the team set amp geometry from sketches and from angles and wheel spacing referenced from other teams; it credits teams 33, 5460 and 3467 for the redesign.

Onshape CAD: Simbot Skyfall 2.0 CAD (Onshape) · CAD release (CD) · Simbot Skyfall 1.0 CAD (Onshape) · Simbot Skyfall 2.0 CAD (Onshape) · Code release (CD) · Code (GitHub)

■ Study pick

3061 Huskie Robotics Artemis

USA-IL

Local EPA 42 (#66)

32-16-2

Pivoting flywheel shooter with kicker; intake with IR note sensing; swerve; 92 lb at Heartland inspection

SHOOTER (SPEAKER)
Kickoff requirements: score in the amp, speaker and trap; adjustable angle greater than 90 degrees; shoot amp and speaker from against the subwoofer. The competition shooter has flywheels on a pivot driven through a gearbox, with a CANcoder after the gearbox to read shooter position. The team added a mechanical hardstop to back up the software pivot limit and changed its plates in the pit at Heartland. At Heartland it also chamfered battery handles that hit the flywheels with the shooter all the way down.

INTAKE
Kickoff requirement: an under-the-bumper intake, double-sided on opposite sides, 'touch it own it'. Week 1 prototypes tested 2 in stealth wheels, a grip-taped hex shaft with a polycarbonate ramp, and sushi rollers; the hex shaft and ramp worked better than the wheels, but the note stuck on the polycarbonate. The competition intake is a removable module with IR sensors for note detection; the team swapped in a spare at Heartland to replace worn wheels. The shooter hardstop and pivot made the swap harder than planned, so the team plans snap-on spacers so the intake can collapse for removal.

FEEDER / AMP / TRAP
Week 1 ideation lifted notes into a central area and flipped them around a drum to stage them for the shooter; whether the final robot uses the drum is unconfirmed. The Heartland recap names a kicker between the intake and the shooter, and an inner rail that guides notes up into the shooter. Notes sometimes stuck on that rail, so the team added a 3D-printed part and later replaced it with a smoother print. The kickoff plan scored amp and trap with the shooter; the final amp and trap method is unconfirmed.

DRIVETRAIN
Kickoff requirement: MK4i swerve drive, as fast as possible, with clearance for notes. The Heartland recap confirms swerve modules with tread, Falcon motors on the modules (the recap mentions soldered Falcon wires during a module swap), and a spare module the team swapped in after a steering motor failed in match 60. The front rail bent during defense and was replaced; the electrical board mounts upside down under a removable bottom polycarbonate panel.

ENDGAME
Kickoff requirement: support the robot on the chain with at least 1 other robot and raise it off the ground. Trap scoring was a shooter requirement. A separate Climber CAD document is linked, and the week 1 climber update links the WCP GreyT Telescope product page. Final climber hardware is unconfirmed.

AUTONOMOUS
3061-lib v2024.0.0 adds a shouldFlipAutoPath method that PathPlanner calls before an auto to flip the path across the field center, and a tuning auto for swerve rotation. In its 2025 build thread the team says it ended the autos of this robot with a sprint and the swerve modules in coast mode, to travel from the centerline toward the source between auto and teleop.

VISION
A Limelight is on the robot. Its first mount let the bumper block a third of the view during intake, so the team printed a new mount at Heartland. 3061-lib v2024.0.0, the base for the code, supports multiple PhotonVision cameras for pose estimation using PhotonVision's MULTI_TAG_PNP_ON_COPROCESSOR feature.

PUBLIC CAD / DESIGN
Full robot CAD (Onshape)

LESSONS FOR STUDENTS

- Each subsystem is a separate Onshape document, so one mechanism can be studied without loading the full robot.
- The team wrote per-subsystem design requirements at kickoff, for example a 2' 3.5" max height, and designed against them.
- When they found no public note friction data, they measured it: each material was tilted under a note until it slipped, and the angle was recorded.
- The Heartland recap pairs each failure with a short-term pit fix and a longer-term fix, for example a spare swerve module with its own CAN IDs for faster swaps.
- A pivot with backlash can still be measured: the team placed the CANcoder after the shooter pivot gearbox and added a mechanical hardstop behind the software limit.
- 3061-lib holds game-agnostic features (AdvantageKit logging and replay, PhotonVision pose estimation, drive-to-pose), and frc-software-2024 holds the game code.

Onshape CAD: Full robot CAD (Onshape) · Build thread (CD) · Full robot CAD (Onshape) · Climber CAD (Onshape) · Drivetrain CAD (Onshape) · Intake CAD (Onshape) · Shooter CAD (Onshape) · Code (GitHub) · 3061-lib v.0.0 release (CD)

■ Study pick

581 Blazing Bulldogs Titan

USA-CA

Local EPA 40 (#78)

46-16

Pivoting split-flywheel shooter; angled amp/trap elevator; chain-driven climb hooks with trap; MK4i swerve

SHOOTER (SPEAKER)
Two Kraken X60 motors power eight 4 in OD polycarbonate flywheels wrapped in CatTongue grip tape. Miniature gearboxes reverse the lower flywheels, so one Kraken X60 drives each side. The left and right sides run independently to put horizontal spin on the NOTE; the flywheels spin on dead axles. One Kraken X60 drives the pivot through 3D printed Onyx herringbone sector gears.

INTAKE
Roller geometry was first modeled on the 6328 V20 intake. One Kraken X60 drives all rollers; geared shafts reverse direction, and 9 mm wide 5 mm HTD belts link the shafts. The first contact roller is 1/2 in fiberglass tube in 7/8 in OD surgical tubing, so it springs back after a hit. Bent 1/16 in HDPE plates and printed parts funnel the NOTE; beam-break sensors confirm the intake.

FEEDER / AMP / TRAP
Two 1 5/8 in flex wheel queuer rollers hold the NOTE. The team pulses them backwards a few times so a squashed NOTE can return to round. Amp and trap scoring use a one-stage angled elevator on one Kraken X60 at 6.25:1, with rollers on a second Kraken X60. Three top rollers dunk the NOTE; for the TRAP the rollers alternate voltage to 'jackhammer' it in.

DRIVETRAIN
Four SDS MK4i swerve modules, L3 gearing with 16t pinions, eight Kraken X60 motors. Frame perimeter 25.5 in x 28.5 in without the intake, 30 in x 28.5 in with it. Cut-outs in the frame let the elevator pass under the STAGE.

ENDGAME
Each hook is two 1/8 in aluminum plates with printed webbing. Bearing blocks ride #25H chain driven by two Kraken X60 at 15.6:1; one output shaft through both gearboxes keeps the hooks parallel. The 5-stage climb runs on two buttons: hooks engage, the amp elevator extends onto the STAGE on wall-ride wheels, hooks retract, the rollers score the TRAP, the amp mechanism partly retracts.

AUTONOMOUS
Paths built in PathPlanner, with autos for each position around the SUBWOOFER. The team states an auto from any position can score six NOTES. If the beam breaks do not see a NOTE, the path skips to the next NOTE instead of shooting.

VISION
One Limelight 3G localizes the robot. Pose estimation sets coarse speaker aim; the TX angle to the speaker tag then aims the drivebase and the TY angle sets shooter angle and RPM. Feeding shots use pose estimation from all field tags. Near the amp-zone corner the robot angles the shot to bounce the NOTE into the speaker.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- A late redesign can work with a fixed schedule: the team states 4 days of design, 7 days of assembly and wiring, and 9 days of debug and practice before its first event.
- Read the trap climb as a mechanism and a sequence together: the amp elevator doubles as a brace against the STAGE, and two buttons drive all 5 stages.
- Friction is a design variable: the team reduced shooter pulley diameters at fixed belt lengths after it saw high current draw. It set intake belt center-to-center distances -.03 short after 6328 suggested it look at intake current draw.
- Miniature reversing gearboxes let one Kraken X60 drive both upper and lower flywheels on each side, and independent sides give NOTE spin.
- Two state machines run teleop: a note manager tracks the NOTE inside the robot, and a robot manager coordinates scoring, climbing and trapping.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video · Robot reveal (CD)

■ Study pick

4795 Eastbots Perseus

USA-NC

Local EPA 39 (#93)

47-13

22 lb pivoting shooter/amp tower on 2x NEO Vortex at 144:1 with gas shocks; intake handoff; MAXSwerve

SHOOTER (SPEAKER) Shooter on a pivoting tower that also scores in the amp. The tower weighs 22 lbs. Two NEO Vortex motors drive the pivot through a 144:1 combined reduction, with two gas shocks for stability. In the January design review the team chose a horizontal-axle shooter, inspired by Team 95, with each side driven independently. The team chose it for better predicted consistency with warped notes and saw no significant benefit from added spin.	**INTAKE** The intake sits outside the swerve perimeter and inside the bumper perimeter. A handoff allows intake in multiple positions. In January the team prototyped an over-the-bumper intake, sketched an 'in-between' intake and did not prototype an under-the-bumper intake. It kept the over-the-bumper and in-between concepts. In a December 2024 post the team wrote that its offseason intake worked 'maybe even better than Perseus's intake'.
FEEDER / AMP / TRAP The intake hands off to the pivoting tower. The same tower scores in the speaker and the amp. In the January design review the team preferred active control of a note until it is scored over passive amp ramps.	**DRIVETRAIN** Swerve. In a December 2024 post the team wrote that it ran MAXSwerve for the past two seasons before moving to WCP Swerve X on its offseason robot. Frame size is 31 in x 26 in.
ENDGAME Unconfirmed	**AUTONOMOUS** In January the team was building a Kitbot to place on a swerve drivebase so programmers could start path work early.
VISION Three AprilTag-detecting cameras; the code gathers and processes their data for robot positioning. The code also has a note tracking pipeline and a 'note relative' drive mode.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Use passive assist on a heavy pivot: two gas shocks support the tower.
- Combine encoders for pivot control: the code interpolates between an absolute and a relative encoder.
- Pick a concept against a stated failure mode: the January review chose a horizontal-axle shooter because the team was concerned that note deformation and wear would vary compression in a vertical-axle shooter.

Onshape CAD: Onshape CAD · CAD, code and binder release post (CD) · Onshape CAD · Code (GitHub) · Tech binder (Google Drive) · Open Alliance build thread (CD)

■ Study pick

6329 The Bucks' Wrath Dial

USA-ME

Local EPA 39 (#95)

53-19-1

High-pivot shooter that also scores amp and climbs; full-width under-bumper intake; polycarb superstructure; MK4i swerve

SHOOTER (SPEAKER) High-pivoted shooting system. The arm length is 12.447 in, set by other features of the arm system in the reference sketch. The lowest 'home' angle is 25 deg; the OP reports this angle limited shot distance and recommends going lower. Arm position uses only the Krakens' internal encoders: the arm starts each match folded down against TPU hard stops and is not re-homed during the match. The top of the shooter is open in the amp position; the team states no retention roller was needed.	**INTAKE** Full-width under-the-bumper intake. The first intake was over the bumper (test video from 2024-01-24); the team says it worked but did not center notes well and was not wide. Most teams it talked with that were going to Granite State used under-bumper intakes, so the team tested one on the same geometry and had it running on 2024-01-27. The 1x1 front tube bent in play. Before the Championship the team filled it with maple, added a 1x1.5 aluminum angle under it, and remade the bumpers in maple.
FEEDER / AMP / TRAP The shooter flips up to score in the amp.	**DRIVETRAIN** SDS MK4i swerve modules with the L3 ratio and standard black nitrile SDS wheels (team post 2024-05-06).
ENDGAME The pivoting shooting system is also used to climb, at various spots on the chain.	**AUTONOMOUS** Unconfirmed
VISION Basic tx/ty tracking on the AprilTag under the speaker opening, used for shooting while moving. No localization.	**PUBLIC CAD / DESIGN** Onshape CAD and tech binder

LESSONS FOR STUDENTS

- Set the low end of a pivot range from the shots you need: the 25 deg home angle ended up limiting shot distance.
- Structure from material on hand: two sandwiched sets of 3/16 in polycarbonate plates with 3/8 in hex standoffs, tested on the prototype with aluminum as the fallback. The OP reports no plate broke or bent all season.
- Fix a bending member at its root: the team filled the bent intake front tube with maple and added an aluminum angle under it.
- 3D printing at scale: PolyMax PLA on a Bambu X1C, 4-6 walls, 30-50% infill and WCP 1/2 in hex inserts, with PC for parts that kept breaking in PLA. Printed herringbone gears run in pairs on each side for load sharing and redundancy.

Onshape CAD: Onshape CAD and tech binder · Reveal and CAD release thread (CD) · Onshape CAD and tech binder · Reveal video · Early over-bumper intake test (video)

■ Study pick

1787 The Flying Circuits Ronnie

USA-OH

Local EPA 38 (#123)

30-11

Pivoting flywheel 'accelerator' (1 Kraken per side); 'snake path' intake; passive 'slam dunk' hood for Amp and Trap

SHOOTER (SPEAKER)

The team calls the shooter an 'accelerator'. It pivots on an arm and has 4 split flywheel axles. In February the team joined the axles in the middle with a 3D-printed part and bearings instead of cantilevering them. Each side is driven 1:1 by one Kraken (team reply, 2024-03-22). The wheels were BaneBots 3-7/8 T81; after week 1 the team planned tests with 4 in AndyMark Stealth Wheels for more compression. For note spin the team changed from one side spinning faster to more wheels on one side. Before week 1 the Speaker shot was tuned from the subwoofer to a little behind the podium. After week 1 a mentor made a Desmos equation for the arm angle that accounts for the note's parabolic path. Side panels are 1/2 in polycarbonate. The off-season Noodle uses painted wood panels and one NEO per flywheel (2 per side).

INTAKE

Ronnie uses a 'snake path' intake that routes notes around to the accelerator and supports Trap scoring. Before week 1 the team reported fast pickup, but the transfer to the accelerator sometimes took longer than desired. At Greater Pittsburgh, notes taken off-center went through folded against one side. The planned fix was to remove the silicone from the center of the rollers. Noodle uses a direct path, and the team reports that notes seem to intake faster.

FEEDER / AMP / TRAP

Amp and Trap use a passive 'slam dunk' hood above the accelerator. The accelerator swings into the Amp or Trap, and the contact pushes the hood over the note exit, which turns the note down. The team added a string-and-pulley system to pull the hood down past a set angle. At Greater Pittsburgh the string came loose in one match and two Amp shots flew upward; the team was working on 3D-printed pulleys for the string as a fix. A note that could not leave the hood also stalled the flywheels and jammed the accelerator for the rest of a match, so the team planned to reduce hood compression. Noodle is a Speaker and Amp only robot.

DRIVETRAIN

Swerve. The team's 2024-01-09 intake discussion refers to the placement of the swerve modules.

ENDGAME

Chain climb on the Stage, with a planned Trap shot. The team wrote that the climber would lift the robot high enough for the hood to face the Trap, so the Trap shot would work like the Amp shot. Before week 1 the robot climbed the chain, but the Trap shot still needed tuning. At Greater Pittsburgh the climb was manual and slow: the driver stepped the arm through setpoints, then moved the hooks. The team found it cannot lower the arm after climbing, or the robot tips forward. Planned changes were a 'quick climb' with the accelerator down, a latch to hold the robot up after the match, a reshaped hood to open the Trap door, and an extra roller at the bottom of the hood.

AUTONOMOUS

Before week 1 the robot ran three different three-note autos, one starting near each face of the subwoofer. After Greater Pittsburgh the team set a goal of at least a 5-piece auto, and a 6-piece auto if possible, for the Buckeye Regional.

VISION

In the 2023 off-season the programmers started custom AprilTag localization software. After week 1 the team planned note tracking, at least for autonomous, to correct paths when a pickup misses.

PUBLIC CAD / DESIGN

Onshape CAD

LESSONS FOR STUDENTS

- A passive mechanism can replace an Amp/Trap actuator: the hood is pushed into place by contact when the accelerator swings into the Amp or Trap.
- Write down failure modes after the first event: the team found a loose hood string and a hood that stalled the flywheels by over-compressing the note, and planned a fix for each.
- Model the note's arc: misses at week 1 led the team to replace a straight-line aim model with a Desmos equation for a parabolic path.
- One requirement shapes the note path: the Trap led to Ronnie's longer 'snake path' intake, and dropping it on Noodle allowed a direct path that the team reports intakes faster.
- Motor and material choices can change between versions of one design; compare the shooter motor and side panels of Ronnie and Noodle.

Onshape CAD: Onshape CAD · Onshape CAD · Open Alliance thread (CD) · Code (GitHub) · Open Alliance thread post (CD) · Off-season reveal 'Noodle 3.0' (CD) · Noodle photos and videos (Google Drive)

■ Study pick

5951 Makers Assemble

Israel

Local EPA 38 (#113)

43-21-2

Fixed-speed shooter; under-bumper intake; pivot arm; no climber; swerve (off-season robot, 2056 concept)

SHOOTER (SPEAKER) Fixed speed; the team adjusts only the angle, from a linear interpolation table. The team widened the space between the polycarbonate shooter walls after notes stuck. At high speed the shooter wheels widened and rubbed a support rod; the team was moving the support. The shooter beam break sensor was unreliable in some lighting.	**INTAKE** Under-the-bumper roller intake, following the 2056 concept. One prototype tested passive centering on one side and active centering on the other; the team kept passive. Roller pulley slack was fixed with full-length shafts.
FEEDER / AMP / TRAP Pivot arm with an 'arm feeder' on sushi rollers; the team respaced the rollers to remove a dead spot. The team prototyped the arm angle for amp scoring. Arm backlash was reduced with oversized shafts in the arm gearbox and Loctite 648. A limit switch was added for consistent arm zeroing.	**DRIVETRAIN** Swerve, 29 in x 31 in frame. On two modules the team lifted both motors, not one, to widen the intake, so the motors stay uniform and easier to work with.
ENDGAME No climber. The team opted out of climbing because of the added complexity and limited strategic impact.	**AUTONOMOUS** Two auto routines running as of 2024-10-16, with a third planned.
VISION Limelight; the team repositioned its stand for a better view. Software includes auto angle alignment for the amp and a 'stop and shoot' feature. The team reported odometry jumps that occasionally caused missed shots.	**PUBLIC CAD / DESIGN** Off-season robot CAD (STEP, Google Drive)

LESSONS FOR STUDENTS

- Build from a published binder: the team used 2056's technical binder and drawing package and scaled the concept to its capabilities.
- Test a design choice with one prototype: passive centering on one side, active on the other; passive worked and was kept.
- Trade module symmetry for intake width: the team lifted the motors on two swerve modules to widen the intake.
- Arm backlash: oversized shafts and Loctite 648 in the arm gearbox, plus a limit switch for consistent arm zeroing.
- Software structure: after the season the team moved to a state machine and an IO layer similar to 6328's, which made it easier to test code without the robot.

Drive folder / doc: Off-season robot CAD (STEP, Google Drive) · Open Alliance thread with off-season build and CAD release (CD) · Off-season robot CAD (STEP, Google Drive) · Open Alliance build thread (CD) · 'MAroon' reveal thread (CD) · Code (GitHub organization)

■ Study pick

5419 Berkelium Radium the Roomba

USA-CA

Local EPA 37 (#132)

26-14-1

Elevator + arm shooter; 4-side ground pickup via street sweepers; Amp and Trap; chain climb; swerve

SHOOTER (SPEAKER) Variable-angle shooter. The arm sets the shot angle and is fixed to the shooter at one angle. The code auto-aims into the Speaker from anywhere in front of the wing line, with full-speed, velocity-compensated auto shoot.	**INTAKE** Under-bumper intake that picks up from the front and back and sits under and in the belly pan, protected from collisions. Compliant star-wheel 'street sweepers' sweep notes toward it, which gives ground pickup from all 4 sides. The team moved the sweepers inward and cut them down to stay inside the frame perimeter, and drove the two sides independently because the poly cord did not transfer torque well. A planned polybelt feeder was dropped as a degree of freedom with minimal return.
FEEDER / AMP / TRAP An elevator and arm carry the shooter. The elevator reaches the Trap, gives a downward angle on the Amp shot for consistency, and lets the robot shoot over defense. The arm is light and fits inside the moving stage of the elevator. The code runs an automated note pickup and handoff sequence.	**DRIVETRAIN** Swerve drive.
ENDGAME Climbs on the chain and scores in the Trap. One-way spring-loaded hooks on mini-carriages let a single elevator stage climb twice as high. A team member confirmed the hooks use the return-side rigging of the elevator. The robot drives under the Stage.	**AUTONOMOUS** The team states 4-side pickup is a key speed feature in auto. Driving the sweeper sides independently lets the robot pass notes under itself for more auto options.
VISION One camera running PhotonVision. Its pose is fused with swerve odometry for field localization.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Intake geometry can widen pickup without a second intake.
- Count the degrees of freedom you add. The team dropped a planned polybelt feeder under the robot because the sweepers already moved notes to the intake.
- The return side of elevator rigging can do work.
- Test intakes with worn game pieces. The team reports that notes sometimes stuck under the robot, especially older used notes.
- The code lets the driver queue one command in advance to cut transition time between states.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Technical binder (Google Drive) · Code (GitHub) · Robot reveal (CD)

■ Study pick

131 CHAOS Double Time

USA-NH

Local EPA 36 (#144)

45-25-1

Launcher scores Speaker, Amp and Trap; under-bumper intake; lift with chain-pull notches for the climb

SHOOTER (SPEAKER) The launcher centers and indexes notes, launches them, and scores in the Amp and Trap. The team designed it to put spin on the note but never used that feature.	**INTAKE** Under-the-bumper intake, packaged in the drive subsystem with the bumpers and electronics. The team first dismissed an under-the-bumper intake. They changed after 95's build thread showed them how much harder an over-the-bumper intake made integration.
FEEDER / AMP / TRAP Indexing, Amp and Trap scoring all happen in the launcher. Changed between Week 0 and Week 1: the outer roller went onto its own motor so the robot could score in the Trap by pushing the note out the top. Before that, one NEO Vortex drove the outer roller and the other rollers. After the team fixed height issues seen at Week 0, it no longer needed the top path. It kept the separate roller because it lets the robot take a note directly from a human player.	**DRIVETRAIN** Unconfirmed
ENDGAME The lift includes notches that pull the Stage chain down to start a climb. The launcher scores in the Trap. In the reveal thread the team states that Week 0 taught them about 'vertical efficiency'.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Adapt a lift design to the parts you can buy. 131 could not get the WCP inline bearing blocks, so they designed for clamping blocks and moved the pulley supports inside the tubing to keep full carriage travel.
- One mechanism can do two jobs: notches on the lift pull the Stage chain down to start the climb.
- Read other teams' build threads during the season. 131 changed to an under-the-bumper intake after 95's build thread showed the integration cost of an over-the-bumper intake.
- Putting one roller on its own motor added a Trap scoring path out the top of the launcher and a human-player intake path.
- The team states that attending Week 0 got them mechanically complete early, which left time for software features that reduced driver workload.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Code (GitHub) · Robot reveal (CD)

■ Study pick

3357 COMETS Vaisala

USA-MI

Local EPA 36 (#150)

47-15

Roller-pinion sector gears; silicone-over-tube rollers; 14 SPARK MAX; 24x28 in, 75 lb swerve

SHOOTER (SPEAKER) Shooter fed by a shared intake/index/shooter system that uses 4 motors. The OP calls the shooter deck probably the team's most dense set of mechanisms. The robot also has a pivot on 1 motor and sector gears with a roller-pinion profile; which joint each drives is unconfirmed. Each motor has a SPARK MAX controller.	**INTAKE** Rollers are polycarbonate or aluminum tubes, in different diameters for intake and indexer, covered in white silicone tubing. The team seats the undersized silicone with the Spectrum method.
FEEDER / AMP / TRAP Indexer rollers use the same silicone-over-tube build. At the reveal the team had several Amp scoring methods and expected to change them during the season; the final Amp method is unconfirmed.	**DRIVETRAIN** Swerve; the team's 2nd competition swerve robot. 8 SPARK MAX controllers run the swerve. 24x28 in frame perimeter, 75 lb with no ballast.
ENDGAME Stage climb driven by 1 motor.	**AUTONOMOUS** Won the Autonomous Award at FIM District Battle Creek.
VISION A camera sits on an upright (named in a reply that the OP answered).	**PUBLIC CAD / DESIGN** GrabCAD CAD (STEP/IGES)

LESSONS FOR STUDENTS

- Design gears for your machine: a roller-pinion profile (from team 3476's 2019 turret) with 3/16 in dowel pins let the team cut sector gears with one 3/16 in router bit, where their 2022 turret gear needed a 1/16 in bit and 3 operations.
- Plan inspection for custom gears: the team checked the polycarbonate gear teeth every match and the chain tensioner about once a day. The teeth cracked when the rotation encoder lost position and the motor tried to run through the lower hard stop.
- Size for speed: the OP states a small, light robot to stay fast and avoid defense.
- Keep the control system small: 14 SPARK MAX controllers in total, with CAN utilization of 50-60% reported by the team.
- Make status LEDs serviceable: the team mounted LED strips in plastic rectangular tube with printed blocks and hot glue, had bad solder joints and connector problems at the strip base, and recommends a printed attachment instead.

GrabCAD: GrabCAD CAD (STEP/IGES) · GrabCAD CAD (STEP/IGES) · Reveal and CAD release thread (CD) · Reveal video · Champs recap video

Profile

4043 NerdHerd Notezart

USA-OR

Local EPA 36 (#145)

46-16

RI3D-style arm with flywheels and kick-up feed wheels; intake in a protective frame box; first-year swerve

SHOOTER (SPEAKER) Flywheel shooter on an arm similar to the Unqualified Quokkas Ri3D design, with some differences. Flywheels are Stealth wheels; one side has TPU socks that add spin to the note.	**INTAKE** The intake sits in a front 'box' frame extension that protects the arm at its lowest position. The intake does not attach directly to the drivebase. The side and front 1x2 tubes are thicker aluminum extrusions than the CAD shows, joined with custom steel L-brackets. The team chose this after testing a mock frame extension early in the season.
FEEDER / AMP / TRAP A bar of kick-up wheels on the arm feeds the note into the flywheels once they are at speed.	**DRIVETRAIN** Swerve; this was the team's first year with swerve. The drivebase follows a public offseason system: a fixed drivebase that each mechanism must fit. That offseason drivebase uses SDS Mk4i modules (team drivebase CAD release, 2024-01-06). The team flipped the swerve module brackets for more ground clearance.
ENDGAME Unconfirmed	**AUTONOMOUS** Autonomous Award at PNW District Wilsonville.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Test the weak point early: a mock frame extension showed the team it needed custom steel L-brackets to stop skewing.
- Keep the intake off the drivebase: the intake does not attach directly to the frame.
- Stage the shot: kick-up wheels hold the note until the flywheels are at speed.
- Read the post with the CAD: the team states the front tubes are thicker than the CAD shows.

Onshape CAD: Onshape CAD · Reveal and CAD/code release (CD) · Onshape CAD · Code (GitHub) · Offseason drivebase CAD release (CD)

Profile	5406 Celt-X Canada	Local EPA 36 (#159) 41-21

Flywheel shooter set by arm angle; roller intake; amp scoring; climber with trap mechanism; SDS Mk4i swerve

SHOOTER (SPEAKER) Flywheel shooter. The auto-shoot criteria include flywheel speed and arm angle, so an arm sets the shot angle. Before Houston the team added auto-shoot: the feeder motor runs when the flywheel is at speed, the arm is at angle, the AprilTag is centered and drive speed is below a threshold. A time-of-flight sensor sits halfway up the shooter.	**INTAKE** Roller intake behind a front cross bar. An upgraded 2x2 cross bar held up at McMaster, but a hit bent a roller 1/4 in behind it. An angled hit bent the side plates, and the intake geartrain risked derailment. Spare intake plates went on before Houston. The kickoff requirements called for a floor intake.
FEEDER / AMP / TRAP A feeder moves the note from the intake to the shooter. The team scored in the Amp. The team reports its trap mechanism worked every time at McMaster when the drive team left time to line up and deploy it. A second time-of-flight sensor lower in the intake and a rear-facing CANdle light give the driver earlier notice of a note.	**DRIVETRAIN** SDS Mk4i swerve with CTRE CANcoders, which the team zeroes with Tuner X. Changed before Houston to NEO Vortex drive motors for more speed, with 8mm x 19mm bearings added at the top of the pinion to reduce cantilevered load on the motor shaft. 3D-printed TPU treads on SDS billet wheels; the version on this robot had solid rims that often delaminated. Gyro changed from a Nav-X to a CTRE Pigeon for the provincial championship after the Nav-X reported wrong angles.
ENDGAME Climber and trap mechanism. The kickoff requirements called for a single-robot climb, plus trap or a reliable High Note.	**AUTONOMOUS** The kickoff target for BetaBot was a 3 to 4+ piece auto with centerline notes. At the provincial championship the team wrote a new auto to fit its partners' autos. Before Houston it added and retuned autos.
VISION AprilTag shot curves for the centered speaker tags (#4, #7) and the off-center tags (#3, #8). At McMaster a code fault made the robot use the off-center tags nearly half the time; the team disabled those curves for playoffs. A note-detection feature gives AI-assisted steering on a driver button; a pre-McMaster post says it was not yet in the autos.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Test a feature against match logs: the team found its tag-selection command used the off-center AprilTag curves nearly half the time, and those shots were worse than with no feature.
- Sensor position sets driver feedback time: a second time-of-flight sensor lower in the intake signals a note earlier than the one halfway up the shooter.
- Plan where collision energy goes: the 2x2 intake cross bar held, but hits then bent a roller and the intake side plates.
- Check shaft loads before a motor swap: for NEO Vortex drive motors on Mk4i modules, the team added a bearing at the top of the pinion.
- Schedule a fallback robot: the team later wrote that AlphaBot's design fell behind, and BetaBot was done too late for much iteration.

Onshape CAD: Onshape CAD · Onshape CAD (Spectrum CAD Collection) · Open Alliance build thread (CD) · Robot Skills List (Google Sheets) · 2025 build thread with retrospective (CD)

■ Study pick

6036 Peninsula Robotics Snoopy

USA-CA

Local EPA 36 (#162)

43-19-1

Turreted speaker shooter on an X-contact bearing with auto-aim

SHOOTER (SPEAKER) Turreted speaker shooter with auto-aim. The turret rides on an X-contact bearing from Lily Bearing (NC065XP0) plus a McMaster-Carr radial bearing; the first bearing arrived about a week and a half after the quote, a later spare took about three weeks. An absolute encoder is used only to zero the turret; during the match the code uses the Kraken's internal encoder. The encoder-to-turret ratio was not a clean multiple ('I think it was 140:44?' per the OP), so the turret must start within about 1/3 of a revolution, with a dashboard readout to confirm it is not wrapped.	**INTAKE** Unconfirmed
FEEDER / AMP / TRAP Unconfirmed	**DRIVETRAIN** Unconfirmed; the reveal OP posted the tag '#SwurretGang'.
ENDGAME Unconfirmed	**AUTONOMOUS** Paths come from a custom trajectory optimization app the team wrote. The captain names a fast 'pickup aim shoot pickup' 4-speaker cycle as a focus, run on sensors and conditionals instead of tuned timings. Autonomous Awards at the Arizona Valley and Hueneme Port Regionals.
VISION Custom AprilTag vision software written by the team; the tech binder has detail. The team stress-tested its coprocessors to choose heatsinks and active cooling, and replaced camera lenses it did not like.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Simulation before hardware: the team ran full turreted auto-aiming teleop cycles within 6 hours of build handoff, mostly tuning. Bring-up rules: clamp every motor command, add software and hardware limits, and graph outputs with commands not sent while moving mechanisms by hand.
- Use sensors and conditionals instead of tuned timings; the captain states this is how the team made its auto cycle fast.
- Custom software has a cost: the team wrote its own trajectory app and AprilTag vision, and the captain states this pays off only if the team can keep it ahead of open-source tools such as Choreo or PhotonVision.
- Turret zeroing can be simple: one absolute encoder zeroes the turret, then the motor's internal encoder takes over. The OP calls a two-encoder Chinese remainder theorem setup 'probably' the cleaner solution and would add a second encoder next time.
- Plan tolerances for outsourced parts: the turret took at least 5 or 6 mechanical iterations. The OP would split the laser-cut ring gear into two 1/8 in sheets, because the single sheet was too thick for accurate tolerances.

Onshape CAD: Onshape CAD · CAD and tech binder release (CD) · Onshape CAD · Software lessons write-up (CD) · Strategy document thread (CD) · Reveal video

■ Study pick

111 WildStang

USA-IL

Local EPA 35 (#180)

22-23

Tilting shooter with note compression; lift on the amp note path; LL3 + Coral camera steers intake alignment; swerve

SHOOTER (SPEAKER) The team built several shooter versions. The prototype had horizontal note compression and no feed system. V2 kept the compression but added feed drag; removing the top feed rollers cut slip and raised exit velocity, with less compression possible on the single-sided feed. A V3 with multiple rows of flywheels was planned for CIR; whether it was built is unconfirmed. The shooter tilts and has a 'subwoofer' preset.	**INTAKE** A floor-facing LL3 with Coral, added for Midwest, commands the swerve strafe to line the intake up with a note. In intake mode the driver controls only throttle. The team added it after drivers had trouble aligning the intake in blind spots near the speaker and stages.
FEEDER / AMP / TRAP Notes move from the intake note path into a separate amp note path. A lift is part of the amp scoring sequence. At CIR a failing 775 motor brush jammed the feed for 3-4 matches, and the motor failed in Q43. Changed for Midwest: one LaserCAN aimed down the feed path was too inaccurate, so the team fitted two LaserCAN sensors used as break beams.	**DRIVETRAIN** Swerve. The team dropped to the EH3 ratio from the upgrade kit and runs NEO Vortex drive motors with 50A current limits. The stock REV over-molded wheels slipped and wore, so the team changed to its own TPU tread on Thrifty Bot billet hubs. Inspection weight was around 103 lbs on 2024-02-23.
ENDGAME The team tested its climb on 2024-02-25 and posted a video.	**AUTONOMOUS** A 6-note center-start auto had its baseline path on 2024-03-10; the team needed to trim about 3 seconds. The note-alignment assist also runs in auto. It turns on after about 75% of each pre-planned path to correct drift.
VISION At CIR the team used PhotonVision on an Orange Pi 5 with Arducam 844 sensors; camera pitch moved with the shooter angle. The team reported exposure 'hunting' and hardware stability problems. Changed for Midwest to two crossed Limelights on top of the lift static stage. From the stage range, the edges of their wide view can see the Amp AprilTags past blocker robots.	**PUBLIC CAD / DESIGN** CAD (Onshape)

LESSONS FOR STUDENTS

- Shooter exit velocity depends on contact time and slip: the team traced a V2 loss to drag from the feed rollers.
- A camera can do the fine alignment: the floor-facing LL3 with Coral controls strafe while the driver controls throttle.
- Sensor layout matters: one LaserCAN measuring note distance was too inaccurate, and two break-beam LaserCANs replaced it.
- Mount localization cameras on fixed structure: moving them off the tilting shooter removed a camera-pitch calculation.
- Event reliability: the CIR review lists a failed feed motor, a loose battery that destroyed a roboRIO, and a power wire pulled from the PDH.

Onshape CAD: CAD (Onshape) · CAD (Onshape) · Build blog, teams 111 and 112 (CD) · Climb test video (YouTube)

■ Study pick

2877 LigerBots

USA-MA

Local EPA 35 (#192)

35-27-1

Shooter pivots on an angled cascade elevator; under-bumper intake; split-height bumpers; winch climber; MK4i swerve

SHOOTER (SPEAKER) The 2024-01-31 CAD overview shows a pivoting shooter on an angled elevator. The pivot is a NEO with an overall ratio of 22.5:1. The team had two shooter designs, planned to prototype both, and made the CAD configurable to either. Side/horizontal design: 4 sets of two 2.875 in BaneBots wheels, one NEO Vortex direct-driving each side. Top/bottom design: dimensions taken from 6328's shooter, 4 sets of three 2.875 in BaneBots wheels on cantilevered hex shafts, 2 NEO Vortex motors with a 1:2 increase. The posts do not state which design was used.	**INTAKE** Under-bumper intake. One roller of 1.375 in BaneBots wheels kicks up the note. Then come 3 polycarbonate rollers covered in silicone tubing and 1 roller of 2 in flex wheels. The team reported good intake test results on 2024-02-09.
FEEDER / AMP / TRAP Feeder: a NEO drives 2 sets of 2 in flex wheels through a 2:1 reduction, with polycord to reverse direction on the opposite side. Angled elevator: 1 NEO at 4:1, cascade rigging with belt on the first stage and rope on the second. It uses AM bearing blocks, 2x1 WCP punched tube and custom CNC gussets. Changed in March: the elevator was rebuilt with upgraded brackets and cross bars.	**DRIVETRAIN** Swerve with MK4i modules at L2 gearing; the team chose L2 'because that is what we have'. NEO drive and NEO steer as of 2024-01-31; the team said it may switch to Falcons. 23 in x 27 in drive base, 29.25 in x 27.5 in frame perimeter. Steel box tube frame rails to resist bending and lower the CG. Split bumpers: the front is low to push notes away, the back is high so the note passes under.
ENDGAME Winch climber, listed as work in progress on 2024-01-31. Two independent NEOs, each with a 15:1 reduction, one per side. The low front bumper gives the chain more room to sit low in the robot.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Bumper height can be a design variable: a high back bumper lets the note reach the under-bumper intake, and a low front bumper gives the stage chain room.
- One configurable Onshape model held two shooter concepts while the team prototyped both.
- The elevator post states motor, ratio, rigging and structure in one list; check each item against the CAD.
- The team states a reason for its frame material: steel box tube rails to resist bending and to lower the center of gravity.

Onshape CAD: Onshape CAD · Onshape CAD (Spectrum) · Open Alliance build thread (CD)

■ Study pick

9312 NERD Spark BATMAN

USA-MI

Local EPA 35 (#187)

33-23

Intake-shooter tilted by a printed CF 4-bar; 2 DOF arm with wrist for amp and chain climb; swerve

SHOOTER (SPEAKER)
Prototype (2024-01-23): four 4 in Thrifty solid urethane wheels (45A) on two NEO Vortex at 1:1 direct drive, open loop at 100% output, with 1/2 in compression. The wheels touch only one side of the note to spin it, at a cost in forward speed. Shot angle about 35 deg, 5-6 in off the floor; the team then began implementing closed-loop velocity PID. On the competition robot a 4-bar tilts the combined intake-shooter; the team wanted higher shots over its back side when close. In a 2024-03-05 post the OP believes the team limits its NEO Vortex motors to 40 amps.

INTAKE
Combined ground intake and shooter, planned to use about 60% of the chassis. Every intake part is Nylon 6 CF. One connecting member of the intake-shooter is designed as the weakest part so it breaks first. The team expects the aluminum hex shafts to fail on a direct hit and planned to cut spares.

FEEDER / AMP / TRAP
Amp: the shooter passes the note into a passive capture mechanism on a 2 DOF arm. A wrist motor with the team's NEO 550 harmonic gearbox sets the note flat against the amp wall (programmed 2024-02-26). At Jackson the transfer relied on gravity to feed the note down; the team reports it was slow and hard to line up.

DRIVETRAIN
Swerve. The 24 in wide by 28 in long chassis was generated from the team's reconfigurable Onshape swerve chassis model; a sponsor laser cut the parts from the exported DXFs. The swerve pods use Thrifty Absolute Magnetic Encoders.

ENDGAME
Chain climb with the arms. Hooks on the forearm drive belt pass the chain to hooks on the back arm, which pull it down to latches. At Jackson the chain hung about 1.5 in higher than the team's practice chain. The team tried to climb in the first 3 matches and stopped, as the arms would break before the climb finished.

AUTONOMOUS
At Jackson one shooting map served both alliance colors in auto and teleop; tuning it for one case broke the other three. The team split red and blue autos and used fixed set points for the 4-bar. By the end of qualifications the robot scored 3 of 4 notes in auto on both colors, and all 4 in the last qualification match.

VISION
A shooting map uses robot pose from vision and odometry to turn the robot and set the 4-bar tilt. The team reports vision and pose estimation issues in teleop at Jackson.

PUBLIC CAD / DESIGN
CAD (Onshape)

LESSONS FOR STUDENTS

- A parametric chassis saves build time: the team set 24 x 28 in, exported DXFs from its reconfigurable Onshape model, and had laser-cut parts back in under a week.
- Tune and test a pose-based shooting map per alliance color and per mode. The team's single map broke for three cases when tuned for one, and set points fixed the autos.
- Check climb reach against field tolerance: a 1.5 in difference in chain height between the practice field and the event stopped the climb.
- Plan where the robot breaks: the team made one intake-shooter member the weakest part and planned to cut spare hex shafts before its first event.
- Measure cycle time on a new mechanism early. The gravity-fed amp transfer worked but was slow; the team writes that 'cool and interesting does not equal effective'.

Onshape CAD: CAD (Onshape) · Build thread (CD) · CAD (Onshape) · Amp wrist harmonic gearbox CAD (Onshape) · NEO 550 harmonic gearbox thread (CD, 2023 offseason) · Shooter prototype video · Robot video, 2024-02-22 · Robot release video, 2024-03-29

■ Study pick

8033 Highlander Robotics Banshee

USA-CA

Local EPA 34 (#208)

33-13

Pivoting shooter on a Maxspline shaft, 2 independent wheel sets; UTB intake; cascade elevator amp mech; Mk4i

SHOOTER (SPEAKER) Two independently controlled sets of Stealth wheels create spin. A Maxspline cross shaft takes virtually all the torsional load, so a chain on one side drives the shooter without twisting or flexing. The same Maxspline is the pivot, riding on a .75 diameter extrusion with needle bearings. A dead-axle roller for the notes rides on top of it through a large thin-section bearing from TTB.	**INTAKE** Under-the-bumper intake with 2 sets of dead-axle rollers; the front one is wrapped in silicone. Two sets of powered 4 in compliant wheels center the note. Pulleys with oversized flanges are used throughout, and the team recommends them on long belt runs such as the centering wheel assembly.
FEEDER / AMP / TRAP Cascade elevator rigged with cord (written 'UHW' in the post) and tensioned with WCP ratchet plates. Two Krakens on MAXPlanetary gearboxes drive it through chain with TTB chain combs. The amp mechanism uses several sets of rollers to deposit the note and hold it laterally. The original geometry could score in the trap with arms, but the team reports this was not reliable.	**DRIVETRAIN** Mk4i modules with Kraken X60s. The team started at L3.5 and moved to L2.5 to push through defense on its 3D printed TPU wheels. The wheel CAD is public; team replies give the print setup (Overture TPU 90a, 0.6 mm nozzle, about 8.5 hours per wheel, about 3 per tournament) and Onyx hubs at about 40% infill.
ENDGAME A servo-actuated latch on the elevator is used for the climb. The trap arms on the amp mechanism carry hooks for a backup climb, which the team reports worked well. Trap scoring was not reliable.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A shaft that takes the torsional load lets a wide shooter be chain-driven from one side without twist.
- Gear the drivetrain for the game: the team moved from L3.5 to L2.5 to push through defense.
- Use pulleys with oversized flanges on long belt runs, as on the centering wheel assembly.
- Robustness has a weight cost: the team reports 124 lb at the Championship and names it as something to avoid in the future.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Code (GitHub) · TPU drive wheel CAD (Onshape)

■ Study pick

341 Miss Daisy Miss Daisy XXII

USA-PA

Local EPA 33 (#247)

38-22-1

Full-width over-bumper intake feeding an under-bumper intake; belts on the swerve modules center notes

SHOOTER (SPEAKER) Unconfirmed	**INTAKE** Full-width over-the-bumper intake that can grab notes at the extreme edges of the robot. An under-the-bumper intake then takes the note. Timing belts wrapped around the intake-side swerve modules pull edge notes to the center before they reach the under-the-bumper intake.
FEEDER / AMP / TRAP The note passes from the intake into a conveyor.	**DRIVETRAIN** Swerve. The team chose belts around the modules over a non-standard drivetrain to center notes.
ENDGAME Unconfirmed	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Two intakes in series: a full-width over-the-bumper stage for reach, then an under-the-bumper stage.
- The centering belts ride around the swerve modules, so the team kept a standard drivetrain and still centered notes picked up at the robot's edges.
- The release pairs CAD with code and a design binder for the same robot, so a student can check one against the others.

Onshape CAD: Onshape CAD · CAD, code and binder release (CD) · Onshape CAD · Code (GitHub) · Design binder (Google Drive)

■ Study pick

2930 Sonic Squirrels Maestro

USA-WA

Local EPA 33 (#238)

29-26-1

Continuous belt-driven elevator; PhotonVision note detection and localization; Choreo autos

SHOOTER (SPEAKER) Unconfirmed	**INTAKE** Unconfirmed
FEEDER / AMP / TRAP Continuous, belt-driven elevator, inspired by 2910's 2018 robot. The belt runs are outside the tube, not inside, for maintenance, repair and making spares. The code has teleop auto-align to both the amp and the trap.	**DRIVETRAIN** The robot uses 1678-style bumper mounts for quick bumper swaps.
ENDGAME The code has teleop auto-align to the trap.	**AUTONOMOUS** Trajectories come from Choreo. A state machine lets the robot make decisions during auto, for example whether to skip a game piece. AdvantageKit handles replay, logging and simulation.
VISION PhotonVision for pose localization. A PhotonVision neural network detects game pieces, and the code uses it to adjust paths on the fly.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Use CAD configurations to communicate intent: the top-level model has one configuration per scoring setpoint and an 'adjustable' configuration for the full range of motion.
- The elevator keeps the belt runs outside the tube for maintenance, repair and making spares.
- The team names a state machine that can skip a game piece in auto, PhotonVision game piece detection and Choreo trajectories as the software work behind its Autonomous Award.
- The team adopted 1678-style bumper mounts for quick bumper swaps.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Reveal thread (CD) · Reveal video

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■ Study pick

930 Mukwonago BEARs

USA-WI

Local EPA 32 (#290)

35-16-1

Turreted shooter driven through a strain-wave gearbox; 3D printed gears; SDS MK4i L3 swerve with Krakens

SHOOTER (SPEAKER) Turreted shooter. The turret is driven through a strain-wave (harmonic) gearbox, sourced by a team mentor; the OP reports zero backlash, no noticeable wear all season, and a smaller package. A hex shaft through the turret carries the turret encoder. Shooter gears are 3D printed; the ratio changed from 2:1 to 1.5:1 after the Wisconsin Regional by increasing gear DP without changing the plates.	**INTAKE** The intake bevel gears are 3D printed; the OP reports no issues with them.
FEEDER / AMP / TRAP Unconfirmed	**DRIVETRAIN** SDS MK4i swerve modules, L3, with Krakens and 16t motor pinions to raise top speed. The OP states this is likely the main factor in the robot's speed. Weight about 113 lb at the Championship, and lighter at Lake Superior.
ENDGAME Unconfirmed	**AUTONOMOUS** Unconfirmed
VISION The robot has 3D printed camera mounts.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A strain-wave gearbox can drive a turret in a small package; the OP reports zero backlash and no noticeable wear over the season.
- A hex shaft on the turret axis gave a direct mount for the encoder, without the separate encoder gearbox the team built in 2020/21.
- Printed gear practice from the thread: the OP recommends about 10 DP, and the team printed with 4 walls and gyroid infill. The OP names wall count as probably the most impactful setting, because the bore is the main failure point.
- Printed gears can absorb a ratio change: the shooter ratio changed after the Wisconsin Regional with new gears on the same plates.
- The failure is documented too: the bore of the motor-driven shooter gear rounded out twice, once with the Kraken shaft profile printed in and once with a hex insert.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video

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■ Study pick

2102 Team Paradox

USA-CA

Local EPA 32 (#262)

19-10

Combined intake/shooter with two roller pairs (shoots from either side); climber; swerve; REV motors only

SHOOTER (SPEAKER) Combined intake/shooter mechanism. Software stows the note in the correct pair of rollers and spins up the other pair, depending on the side the robot shoots from. In speaker mode the robot aims passively to shoot or pass while it holds a note. REV motors only.	**INTAKE** Part of the combined intake/shooter. The intake retracts automatically when a note is acquired. A 12-pin Deutsch connector and an Anderson block connect the chassis to the pivot, so the mechanism can be hot-swapped.
FEEDER / AMP / TRAP No separate indexer is described; the note is stowed in the rollers of the combined intake/shooter. An operator switch toggles between amp and speaker modes.	**DRIVETRAIN** Swerve. All wires of one type for a module share one connector, so motors cannot be plugged in wrong during a module swap.
ENDGAME A climber, run from one operator button, with its own master disconnect point for hot-swapping.	**AUTONOMOUS** Unconfirmed
VISION Team-written AprilTag detection software, released on GitHub as the software used in the season.	**PUBLIC CAD / DESIGN** Robot CAD (Onshape)

LESSONS FOR STUDENTS

- Trunk/branch CAN: each CAN device sits on its own short branch, so one bad connection does not take down the whole bus. The team states FIRST does not recommend this layout and that stub lengths must stay short.
- Plan wire paths in CAD: the team sketches wire routes and zip-tie holes in Onshape before wiring anything.
- Master disconnects: all wires for a hot-swappable mechanism pass through one or two connectors.
- Automation to cut driver load: by the team's own estimate, about 80% of matches ran on 4 buttons besides the drive sticks. The team also states the code was hard to debug and does not recommend it as a reference.
- Prototype early: a reused prototyping robot let the team run full cycles by the Wednesday after kickoff and test the intake/shooter combination.

Onshape CAD: Robot CAD (Onshape) · Open Alliance thread with write-ups (CD) · Robot CAD (Onshape) · Code (GitHub) · AprilTag software (GitHub) · Reveal thread (CD)

Profile

7312 T3

USA-TX

Local EPA 32 (#293)

37-22

Fixed-angle two-sided shooter, rotary amp bar, OTB rotary intake, GreyT telescoping climber, Mk4i (Jan 2024 design)

SHOOTER (SPEAKER) Two-sided shooter with ION grip wheels, driven by two separate NEOs at 1.5:1. The note is compressed to 7 in wide. Fixed angle, per the selected Concept 2. Values are from the 2024-01-19 design update.	**INTAKE** Over-the-bumper rotary intake, changed from an under-the-bumper design in week 2. Sets of 2 in Thrifty squish wheels are driven by a Falcon (the post says 'maybe Kraken'), and the intake pivots on a NEO geared 40:1. Three hex shafts near the end of the intake add rigidity and guide the note.
FEEDER / AMP / TRAP Amp: a rotary bar works with the fixed-angle shooter. The team's testing found an upward shot stayed in the amp when a piece of box tube sat about halfway up the opening. Trap was not pursued; the team judged it unattainable.	**DRIVETRAIN** Mk4i swerve, listed as a common part of both January concepts.
ENDGAME Stage climb on GreyT telescoping tubes at 17.5:1. The hook design was still in progress on 2024-01-19. The team ranked ONSTAGE above the amp because the ranking point needs two robots ONSTAGE. Trap not pursued.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Full robot assembly (Onshape)

LESSONS FOR STUDENTS

- A week 1 priority list: speaker, then ONSTAGE, then amp, with trap dropped as unattainable.
- Compare the shooter numbers against other teams.
- The intake moved from under the bumper to over the bumper during week 2 of design; the CAD shows the result.

Onshape CAD: Full robot assembly (Onshape) · Build thread (CD) · Full robot assembly (Onshape) · Mechanisms CAD (Onshape)

Profile	1710 The Ravonics Revolution ADONIS USA-KS	Local EPA 31 (#316) 29-13

Wrist and shooter; intake; rope-rigged elevator; Stage chain climb; swerve

SHOOTER (SPEAKER) The robot has a wrist and a shooter; the team's preseason programming lessons use both. At the Daly Division the team fed notes across the field to alliance partners; its match review states it was not confident passing from the middle of the field.	**INTAKE** The intake frame uses 1x1 in steel tubing bolted to the robot frame, with a Delrin block locking system. At Cow Town Throwdown the tubing bent, the bolt joining the intake bar to the frame sheared, and the Delrin lock failed; the team added rivets to hold the intake. The team's programming lesson for this intake uses beam breaks to sense the note.
FEEDER / AMP / TRAP An elevator rigged with ropes on the center and left side. The team reports rope fraying and snapping all season; at Cow Town Throwdown the left rope snapped and the elevator did not work for the playoffs. A polycarbonate hard stop also broke.	**DRIVETRAIN** Swerve. The Cow Town Throwdown post names the rear 2x1 in swerve tubing. After the season the team kept the ADONIS chassis as a testbed named ALPHA. The team states it decided to change to L3 gearing; the chassis has Kraken drive and Falcon steer motors.
ENDGAME Stage chain climb. The team's Daly Division match review describes the robot climbing at center stage in Qualification Match 120.	**AUTONOMOUS** 4-note auto that scores the close notes. In Daly Division Qualification Match 120 the robot used note detection to pick up a note passed by 118 during auto and score it in the speaker. A preseason programming lesson used PathPlanner.
VISION PhotonVision, with an AprilTag camera and a note-detection camera.	**PUBLIC CAD / DESIGN** Fusion 360 CAD

LESSONS FOR STUDENTS

- Log failures by subsystem after each event. The Cow Town Throwdown post lists each failure and whether the team fixed it or left it as not critical.
- Note detection can earn points in auto beyond the planned path: in a Daly Division match the robot picked up and scored a note passed by an alliance partner.
- Rope rigging on an elevator is a wear item: one rope snap took the elevator out for a full playoff.
- A retired competition chassis can serve as a software testbed. The team kept the ADONIS swerve base as ALPHA to test a new AdvantageKit swerve template before the 2025 kickoff.
- The team states its recent robots weighed 120-125 pounds without bumpers and battery, and set a lighter target for 2025.

Fusion 360 CAD: Fusion 360 CAD · Fusion 360 CAD competition robot) · Open Alliance build thread (CD) · 2025 build thread with robot offseason notes (CD) · Scouting data, Daly Division (team site)

■ Study pick

1155 SciBorgs Taiwanese Fried Chicken

USA-NY

Local EPA 30 (#334)

18-9

Pivoting flywheel shooter that also scores Amp; shooting on the move from pose estimation; swerve

SHOOTER (SPEAKER) Flywheel shooter on a pivot. The team re-tuned the pivot and shooter PID after NYC. The team simulated shot trajectories during development and wrote a public shooting-on-the-move command.	**INTAKE** Roller intake with beam-break sensors. The team moved the beam breaks from soldered joints to Molex SL connectors; a swap then took about 3 minutes instead of about 30. An intake roller shaft failed after a hard hit at RoboReplay and was replaced.
FEEDER / AMP / TRAP A conveyor feeds the shooter. Amp scoring runs through the shooter; 3D-printed shields around the flywheels were added to prevent note jams while amping. The team reports an inconsistent Amp angle at Duel on the Delaware.	**DRIVETRAIN** Swerve. The team's SOTM talk covers shooting on the move with a swerve drive.
ENDGAME Stage chain climb with climb-lock hooks and climb plates. One hook broke in a RoboReplay qualification match and the robot still climbed.	**AUTONOMOUS** A center 5-note auto ran at RoboReplay, where the team won the Autonomous Award. The team says its autos were weaker earlier in the season.
VISION Cameras feed pose estimation used for shooting. The team's running theory is that a slowly loosening camera mount degraded pose estimation and shot accuracy. It found and fixed the mount at RoboReplay. The robot also carries an Orange Pi.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Check sensor mounts when accuracy drifts: the team first suspected pivot slop, then found a loosening camera mount that it believes caused falling shot accuracy.
- Put connectors on sensors that get hit: moving the intake beam breaks to Molex SL connectors cut a swap from about 30 minutes to about 3.
- SOTM is documented three ways: a build-thread post on shot simulation, a talk, and the public Shooting.java command.
- Guard the places where notes jam: the team added 3D-printed shields around the flywheels to prevent jams when scoring in the Amp.

Onshape CAD: Onshape CAD · Onshape CAD #1 · Onshape CAD #2 · Shooting command code (GitHub) · Build thread, SOTM post (CD) · Open Alliance thread with off-season notes (CD)

Profile	3136 O.R.C.A. JT USA-VA	Local EPA 30 (#367) 38-23-1

Arm-mounted intake shoots out the back with two flywheels; AprilTag distance sets arm angle and speed; 24x24 in frame

SHOOTER (SPEAKER)
The intake is also the shooter: it mounts on arms, takes notes in with 4 sets of wheels at the front, and shoots them out the back with two flywheels for both Speaker and Amp. After compression tests the team chose 4 in green compliant wheels. In the first build weekend the team built V1 and V2 of the intake and designed V3. Changed in late March 2024: the log refers to a new shooter with more power, aimed by AprilTag distance. The team reports accurate shots from near the Speaker to about mid-Stage, falling off past the wing. Shooting on the move was tried and dropped for time. A lob shot from outside the opponent's wing to the team's amp area fed notes to partners.

INTAKE
Wide intake with 4 sets of wheels at the front; notes pass through to the flywheels at the back. Early tests found 1 in of compression works well for notes. Changed in late March 2024: the team redesigned and built a new intake after prototyping. At the District Championship a hit cracked the intake; the team patched it with washers and polycarbonate scraps for the rest of the playoffs.

FEEDER / AMP / TRAP
An arm carries the intake and shooter. The arm attachment used a temporary sprocket of about 66T, with a planned change to 68T. In late March 2024 the arm moved to a trapezoidal motion profile for smoother, more accurate aiming. Amp scoring uses the same mechanism; the Trap was not a design goal.

DRIVETRAIN
24x24 in frame. The build log mentions rebuilding swerve modules during build season.

ENDGAME
Chain climb was a kickoff goal with no design chosen at the time; the Trap was not a design goal.

AUTONOMOUS
Kickoff goals were a Speaker shot and a taxi in auto, with no Amp in auto. Students began learning PathPlanner during build. Before the District Championship the team ran a 2 note center auto and a 3 note auto that shoot from the pickup spot. Longer paths drifted left past mid-field, so the team switched to Choreo with help from team 1599. The final autos were an amp-side 3 note and a center 4 note. Won the Autonomous Award at the CHS District Glen Allen VA Event.

VISION
Limelight cameras. Added in late March 2024: the distance to the AprilTags sets the arm angle and shooter speed. FMS issues blocked the Limelights in the team's first District Championship match.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- Sketch a concept in CAD before you build it: the team dropped a double-wristed arm after the CAD sketch showed it was not viable.
- Learn from Ri3D: the chosen layout is close to the Penn State Ri3D robot but shoots out the back of the intake, and the team found it matched the Unqualified Quokkas Ri3D design.
- Test compression early: tests led to about 1 in of intake compression and 4 in green compliant shooter wheels.
- Secure the roboRIO and its power wire: a loose roboRIO and a loose power wire end shut the robot down in both District Championship finals matches.

Onshape CAD: Onshape CAD · Onshape CAD · Code (GitHub) · Open Alliance build log (CD) · Build logs (Google Drive)

■ Study pick

3729 The Raiders Mr. Mimlitz

USA-CO

Local EPA 30 (#328)

15-13-1

Speaker and amp; hybrid under-bumper intake with slapdown and printed funnels; Stage hang; swerve; 5-note auto

SHOOTER (SPEAKER) Scores in the SPEAKER.	**INTAKE** Hybrid intake: a 10.5 in wide under-the-bumper intake, a 'slapdown' that pulls notes in over the bumper, and two passive PETG funnels below the robot to center the note. Funnels were printed at 100% infill on Bambu X1Cs. Slapdown moving plates began as 1/8 in aluminum and changed to 1/4 in Delrin after a collision at a scrimmage. Static plates bent at Central Missouri and changed from 1/8 in to 1/4 in. A Delrin plate broke in CODE Q40; the pit crew repaired it before the next match.
FEEDER / AMP / TRAP The robot places notes in the AMP.	**DRIVETRAIN** Swerve. The frame perimeter is 28 in on the intaking side; swerve modules and cross-rails take 17.5 in of that.
ENDGAME Hangs on the STAGE.	**AUTONOMOUS** 5-note auto. The team chose a wide pickup partly for a faster center-line auto, because the robot travels less to reach each note.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Budget the frame width first: swerve and cross-rails took 17.5 in of a 28 in side, which left 10.5 in for the intake.
- Tie intake width to strategy: the team wanted a wider pickup for driver visibility and a faster center-line auto.
- Plate material follows failures: 1/8 in aluminum moving plates became 1/4 in Delrin, and 1/8 in static plates became 1/4 in.
- A helper can fail without stopping the robot: after a slapdown plate broke in CODE Q40, the robot still picked up notes for the rest of the match.
- Write the retrospective: the team states a wider 95-style under-bumper intake, with two fewer motors, would have suited it better.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Reveal (CD)

Profile	5431 Titan Robotics Orpheus USA-TX	Local EPA 30 (#341) 22-23

V2 rack-and-pinion shooter angler (about 35-55 deg); pivot intake scores Amp; chain-driven Vortex climber; swerve

SHOOTER (SPEAKER)
V1 was a static shooter set at 55 deg, which prevented Amp and Trap scoring. V2 adds an 'angler': a curved rack and pinion, about concentric to the intake pivot, moves the shooter about 20 deg, from about 35 to about 55 deg relative to the note path. The shooter rides on flangeless bearings on eccentric hex shafts that run along a curved 'rib' track; the team went through three iterations of this layout. In January tests the bottom wheels needed 20 to 30% less power than the top wheels to reach the Speaker. A January prototype design whose wheels tore into the note was scrapped; the post does not say if it was an intake or shooter prototype. On 2024-03-09 the team replaced the worst shooter wheels and planned to replace a churro shaft with hex shafts. A revised shooter design was in progress on 2024-03-13.

INTAKE
Pivoting intake, part of an intake-shooter combo. Rollers are polycarbonate tubes wrapped in silicone tubing, with a MAXPlanetary gearbox and small polycord pulleys. V2 uses two NEO 550s instead of one; the team states this made Amp scoring more consistent without overloading the single V1 NEO 550. V2 adds beam break sensors, a protective panel and bearings that let the intake glide on the carpet. The V2 pivot moves the pulleys and belts inside the base plates, because the swerve modules blocked repairs on V1. This stage gives a 2:1 torque reduction. In January, after REV recalled their motor controllers, the backup plan was a NEO for the pivot and a NEO 550 for the rollers; the reduction changed from 15:1 to 5:1 and the NEO 550 then would not run.

FEEDER / AMP / TRAP
The Amp is scored with the intake: the 2024-03-04 practice notes (posted 2024-03-29) describe an auto command that outtakes the intake to score in the Amp. The drive team practiced Amp and Trap scoring on 2024-02-21.

DRIVETRAIN
Swerve with a 29x29 in frame perimeter (V1 plan). On V2, cross beams set the spacing of the intake base plates and shooter plates and stiffen a belly pan drilled with nearly 300 holes on a 1 in pattern for electronics on its underside. A polycarbonate skid plate covers the electronics. The team used CTRE swerve code with Phoenix 6, traced a clicking sound in the modules to a bearing problem, and almost fixed a math issue that stopped the robot at 45 deg joystick positions.

ENDGAME
The V1 plan was an elevator climber. The V2 climber is chain driven by NEO Vortex motors: a chain bolted to the traveling arm runs between a top sprocket and a drive sprocket on a 1/2 in hex shaft. Bearing blocks act as hard stops, and the team rejected an earlier pulley design as too fragile. Retracted, the robot fits under the Stage and chain. On 2024-03-06 the climber ratio changed from 27:1 to 9:1 and pegs were added to the hooks. The right arm's chain was loose and skipped, the climber broke on 2024-03-09, and on 2024-03-14 the team removed an arm while a new climber was designed.

AUTONOMOUS
On 2024-03-06 the team confirmed a 1-piece preload Amp auto. A PathPlanner 2024.1.2 path ending interrupted the Amp outtake command; the mentor's fix was to put the event in the auto, not in the path. The code uses PathPlanner.

VISION
The team calculated a Limelight 3 mount angle of 45 deg, up from a first estimate of 22 deg, for close-range AprilTag accuracy. The team also calibrated PhotonVision. The leads concluded a coprocessor would help with more cameras and that a standalone camera might replace the Limelight. In February the team confirmed an Orange Pi worked but was laggy, and suspected the ethernet cables.

PUBLIC CAD / DESIGN
Fusion 360 CAD (V2)

LESSONS FOR STUDENTS

- Add a degree of freedom when a fixed design blocks a task: the static V1 shooter could not score the Amp or Trap, so V2 added an angler.
- Design for repair: V1 intake pivot belts sat outside the base plates where the swerve modules blocked access, so V2 moved them inside.
- Shot shape can be tuned with wheel speed: the bottom wheels needed less power than the top wheels to reach the Speaker.
- Calculate a camera mount angle before you build. The team's calculations changed the Limelight angle from a first estimate of 22 deg to 45 deg for close-range accuracy.
- Plan for supply problems. When REV recalled their motor controllers, the team made a backup plan with a NEO and a NEO 550 on the intake.

Fusion 360 CAD: Fusion 360 CAD (V2) · Fusion 360 CAD (V2) · Fusion 360 CAD (V1) · Open Alliance thread (CD)

■ Study pick

1414 IHOT Encore

USA-GA

Local EPA 29 (#420)

41-21

Orbit-inspired layout: differential speed shooter; combined intake/indexer; amp deflector; no climb; REV swerve

SHOOTER (SPEAKER) Differential speed shooter. The shooter pivots. Before the anti-jammer, the robot could intake from 0 deg (shooter flat down) up to 70 deg, and moved past 70 deg only to score in the amp.	**INTAKE** The intake and indexer are one combined system. Rollers are mostly dead axle: the axle slides through the side plate, so the plate needs no bearing. Roller stock is 1.125 in ID x 1.25 in OD with 0.0625 in silicone tubing. Pulleys are 34t 3mm GT2 with 0.500 in ID x 0.875 in OD radial bearings (WCP-0777) inside.
FEEDER / AMP / TRAP An 'amp deflector' moves into the path of the note to score in the amp without the precise controls Orbit used. For Championship the team added an 'anti-jammer': two plates with a roller and belts. Before it, scoring in the amp with a note lying on the floor in front of the robot caused a jam. With it, the robot can intake at any shooter angle.	**DRIVETRAIN** Swerve with REV swerve modules. The OP states the REV modules were the only mechanical failure of the season.
ENDGAME No climb and no trap. The OP states both were omitted mainly due to time, because the design started after Orbit's release video.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Copying a published layout still leaves decisions: the team replaced live-axle rollers and bearing plates with dead axles that slide through the plate, to match what it could machine.
- A moving deflector can replace control precision: the 'amp deflector' moves into the note path so amp scoring does not depend on exact shooter control.
- Shared intake and indexer paths create interactions: amping with a note lying on the floor caused a jam until the team added an 'anti-jammer' roller.
- Scope cuts are a design choice: the team omitted climb and trap for time after starting a late redesign.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD

■ Study pick

6647 PrepaTec - VOLTEC

Mexico

Local EPA 29 (#415)

34-21-1

Kraken shooter and intake; elevator removed mid-season; swerve

SHOOTER (SPEAKER) The team prototyped shooter wheels of several types for compression and grip on the note. At Monterrey the shooter sat on top of the elevator; the team states this affected its shots. Shooter motors changed from NEO to Kraken during the season. The pivot used gears; after Hermosillo the team planned a chain pivot to cut backlash and a change to add spin.	**INTAKE** The team describes a touch-it-own-it intake that picks notes from different angles. To close the intake, the code applies equal output to both of its motors in the same direction. Intake motors changed from NEO to Kraken during the season. Note detection used current limits and a debouncer at first; changed at Monterrey to a 50cm beam break.
FEEDER / AMP / TRAP The robot scored in the amp: at Hermosillo it placed two amp notes, then one speaker note while amplified. After Hermosillo the team planned to test the intake as the storage mechanism. The team lists trap scoring as not always needed.	**DRIVETRAIN** Swerve. Drive motors were Falcon 500 at the start of the season; changed before Hermosillo to Kraken motors. The swerve code is the team's AndromedaLib library, moved to AdvantageKit in January.
ENDGAME At Monterrey an elevator formed part of the main structure. Before Hermosillo the team removed it. The team states it almost never climbed in quals and that climbing was not needed in playoffs. After Hermosillo it planned a climber with an improved structure.	**AUTONOMOUS** No working autos at Monterrey. For Hermosillo the team wrote a 3 piece auto and an amp auto that places one or two notes in the amp. Paths use PathPlanner; a note-align command runs at the end of each path. The team planned a move to Choreo after Hermosillo.
VISION Limelight 3 running a team-trained neural network for note detection, trained on over 53,000 pictures. Used mainly in auto to align the robot to a note and start the intake. Localization ran on LimelightOS; after Hermosillo the team planned a move to PhotonVision and more than one odometry camera.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A mid-season simplification with stated reasons: the elevator broke easily, carried the shooter, and served a climb the team almost never used in quals.
- A short auto can set match strategy: the team used an amp auto to amplify early and reports a +20 point lead in its playoff matches.
- Note detection by motor current was unstable on the field for this team; it changed to a 50cm beam break at its first event.
- A robot-wide SuperStructure state machine, inspired by 254 and 1678, conflicted with the mechanisms at Monterrey. The rebuild added end conditions to all commands.
- The thread gives a full SysId routine for swerve, logged with the WPILib callback to a FAT32 USB drive on the roboRIO.

Onshape CAD: Onshape CAD · Onshape CAD · Build thread (CD) · Code (GitHub) · Note-align command (GitHub)

■ Study pick

2491 NoMythic Eurydice

USA-MN

Local EPA 27 (#475)

17-8

Bottom-pivot two-roller shooter that also scores amp; under-bumper intake; telescoping climb; 26x26 in swerve

SHOOTER (SPEAKER) Bottom-pivot shooter with a 'sandwich' of two rollers, one on top and one on the bottom (January 19 concept post). The same shooter scores the amp. At Northern Lights the amp shot needed the robot an inch or two back from the amp, good side-to-side alignment and an extreme shooter angle. For 10,000 Lakes the team planned a button that backs the robot up, angles the shooter and scores on its own.	**INTAKE** Under-the-bumper ground intake with two horizontal rollers. Prototypes showed it can take notes from the front and back, and compliant rollers intook faster than rigid ones. Changed in March after Northern Lights: bottom roller gearing went from 3 times to 2 times drivetrain speed for more force and fewer stalls. Alignment bars pushed notes away, so the team replaced them with ramps under the drivetrain.
FEEDER / AMP / TRAP Indexer design was left open in the January concept. Notes sometimes jammed at the hand-off from intake to indexer, so in March the team added powered side wheels to push the note through. Amp scoring uses the shooter.	**DRIVETRAIN** 26 in x 26 in swerve, per the January 19 concept post. The team anodized its swerve modules purple in the 2023 offseason. The team's V3 wheel guards came through Northern Lights with one small dent.
ENDGAME Two telescoping tube climber arms, one on each side, with self-centering hooks that align on the chain (January concept). At Northern Lights the climbers worked but the climb was slow, and the 1/4 in aluminum hooks bent side to side. The team prepared spare hooks.	**AUTONOMOUS** The team set a goal of a four-note auto for 10,000 Lakes: one preload plus three collected notes, and several autos to fit different alliance partners. Whether these ran at the event is unconfirmed.
VISION A Limelight was used for object detection, per the team's 2025 build thread. Odometry-based shooting worked the night before Northern Lights and failed at the event. The team then added set shots: the driver goes to a known point such as the podium and the robot shoots with hard-coded, pre-tested yaw and pitch. The drivers rarely used this option, so long shots were taken only when needed. Goals for 10,000 Lakes included automatic speaker aim and shooting while moving.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Intake roller speed is a trade against force: the team geared its bottom rollers down so they stall on notes less often.
- Plan a fallback for vision: when odometry shooting failed at the event, the team added shots from known points. The drivers had little practice with them and rarely used them.
- Support the end of a small motor pinion: the team sheared NEO 550 shafts and added a bearing at the end of the pinion to stop the cantilever load.
- Amp shots from a pivot shooter are sensitive to distance; the team planned a button that backs the robot off the amp a fixed amount before scoring.
- Check climber hooks for side loads: the aluminum hooks bent side to side at the first event.

Onshape CAD: Onshape CAD · Onshape CAD · Open Alliance build thread (CD) · Reveal 'Eurydice' (CD) · Reveal video

■ Study pick

7525 Pioneers

USA-TN

Local EPA 27 (#523)

22-13-1

Static shooter; beater-bar intake; roller amp mechanism added for Worlds; two rope climbers; SDS MK4i swerve

SHOOTER (SPEAKER) Static (fixed) shooter. At Arkansas notes sometimes caught on a small lip when shooting; the team trimmed the lip further and reports no more stuck notes.	**INTAKE** Roller intake with a beater bar in front to take hits and protect the rollers. The robot senses a note by motor current. At Arkansas the bar bent and rubbed the rollers, and tape in the gears stopped intaking in the last playoff match. At Buckeye a hit bent the 3/4 in beater tube; the team changed it to 1/2 in hex shaft and added an operator button to turn off the current sensing.
FEEDER / AMP / TRAP Amp mechanism. At Arkansas and Buckeye it was a round tube that flipped up to keep the note from bouncing out of the amp; the team stopped amp shots at Arkansas. Changed for Worlds to a 118-inspired design: a hex shaft with 2 in flex wheels and 3 polycarbonate rollers that bend the note down into the amp. The pivot uses 2 NEOs on a 56:1 reduction and the rollers use a Kraken at 2:1. The team reports 90-95% consistency in testing.	**DRIVETRAIN** Swerve on SDS MK4i modules with SPARK MAX controllers near the modules. In October 2023 the team assembled a new set of MK4i modules and described them as NEO modules. Changed before Worlds to 4 Kraken X60 drive motors and the SDS L3+ upgrade kit; the team reports a speed change from about 14.5 fps to 18 fps on the field.
ENDGAME Two climber arms that use Dyneema rope, zeroed by current sensing. At Arkansas one rope snapped in the first playoff match, the third snap on the same side; the team suspects an abrasive point inside the climber. The left arm's current-sensed zeroing also failed at random points.	**AUTONOMOUS** The team worked with PathPlanner in preseason and won the Autonomous Award at the Arkansas Regional.
VISION AprilTag vision only, with no note detection; the robot drove to pose for automatic scoring (2024-09-22 recap). In preseason the controls team worked on Limelight AprilTag tracking for aligning in auto and while scoring. After Arkansas the team listed AprilTag alignment for the amp as a planned fix.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Log each event failure with a short-term and a long-term fix. The team's Arkansas and Buckeye posts list rope snaps, sheared bumper bolts, a bent beater bar and a breaker hit by loose wires.
- An amp mechanism must suit the shooter that feeds it. The team found a flip-up tube inconsistent with its static shooter.
- Sequence in code can fix a mechanism. The team took the note into the amp mechanism first, then lowered the angle, then scored, and added side standoffs for alignment.
- Give the driver an override for any sensor that can fail. After the beater bar bent and set off the intake current sensing, the team added a button to turn the sensing off.
- A mid-season drive upgrade can be scoped and measured; the team reports the speed change on the field.

Onshape CAD: Onshape CAD · Onshape CAD · Open Alliance build thread (CD)

Profile	1038 Lakota Robotics Vio-Linda USA-OH	Local EPA 25 (#663) 14-16

Flywheel launchers and a 'TRAG' amp/trap scorer that feeds them; under-bumper intake; chain climb; REV MAXSwerve

SHOOTER (SPEAKER) Flywheel launchers for the speaker. The 'TRAG' rollers reverse direction to feed the note into them.	**INTAKE** Under-the-bumper intake. A 3D-printed transition mechanism moves the note from the intake to the TRAG.
FEEDER / AMP / TRAP The 'TRAG' scores in the amp and the trap. It rides on rack-and-pinion-like vertical slides with metal sprockets. Side pulleys act as rollers to orient the note for drop-off.	**DRIVETRAIN** REV MAXSwerve modules with a frame of REV ION channels, the same base approach as the team's 2023 robot.
ENDGAME A lifting mechanism climbs onto the stage chain, and the robot scores in the trap.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- One mechanism serves amp, trap and speaker feed: the TRAG scores the amp and trap, and reverses its rollers to feed the flywheels.
- A 3D-printed transition can bridge an under-the-bumper intake and an upper scorer.
- Reuse a known drive base: the team ran REV MAXSwerve on REV ION channels in 2023 and again on this robot.
- CAD and Java code are both public, so a student can trace each mechanism from CAD to code.

Onshape CAD: Onshape CAD · Onshape CAD · Code (GitHub) · CAD and code release thread (CD)

Profile	3749 Team Optix USA-CA	Local EPA 25 (#622) 17-9

Single-jointed arm carrying a four-bar intake that also shoots; two Limelights for pose-based aim; SDS MK4 swerve

SHOOTER (SPEAKER) The final design from build week 2 is a single-jointed arm carrying a four-bar intake that also acts as the shooter. The arm angle for each shot comes from the robot's field pose. At the San Diego Regional the team reports two runs of 5 matches with no missed Speaker shot from anywhere closer than the podium. The team gives about 95% accuracy inside a marked zone.	**INTAKE** Four-bar intake with polycarbonate rollers, turned by a REV NEO 1.1 on a SPARK MAX. In build week 1 the intake goal was to score in the Amp and pass notes to the shooter. At the San Diego Regional the intake motor oscillated on practice day and acetal parts gave out; the team fixed it. In Qualification Match 76 the intake NEO 1.1 and its SPARK MAX failed, and the team replaced both.
FEEDER / AMP / TRAP The intake also acts as the shooter, so notes pass from intake to shot in one mechanism. The build week 1 intake goal included Amp scoring. Amp and Trap method in the final robot unconfirmed.	**DRIVETRAIN** Swerve with SDS MK4 modules, assembled in build week 1. 3D-printed drivetrain corners, inspired by 3847, extend the frame perimeter by 1/8 in and give fastener tolerance. Bumper mounts use a 1/4-20 bolt that slides into a slotted plate (3847/1678 inspired). The drivetrain uses SPARK MAX controllers.
ENDGAME Build week 1 climb concepts were an angled elevator and a shooter-elevator combination, with Penn State's Ri3D sliding design as a separate idea. The final climb is unconfirmed.	**AUTONOMOUS** A 4 note auto was a priority after the Ventura County Regional and ran at the San Diego Regional. The team says it was often interrupted in qualifications by partners who wanted the same start position.
VISION Two Limelights give field-pose estimation, with a Pigeon 2.0 gyro added for the San Diego Regional. The pose sets the arm angle for each shot and auto-aligns the robot to the Speaker.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Write failure reports: the team logs match, issue, cause, pre-match factors and solution, and plans to inspect spare motors and controllers before events.
- Check the VRM fuse: the team ran the VRM on a 20A fuse until a rule R617 violation forced a change to 10A during playoffs at the San Diego Regional.
- Aim from pose: two Limelights and a Pigeon 2.0 give the field pose that sets the arm angle and the robot heading for each shot.
- Print drivetrain corners: 3D-printed corners add 1/8 in to the frame perimeter and give tolerance for fasteners.
- Set goals before designs: at kickoff the team writes a need, want, wish list with no designs allowed.

Onshape CAD: Onshape CAD · Onshape CAD · Build thread (CD) · Kickoff need, want, wish list (Google Sheets)

■ Study pick

6838 X-SHARC Blackout

Turkiye

Local EPA 25 (#660)

15-11

Shooter on a pivoting arm; touch-and-grab ground intake; amp shot backward from the feeder; swerve

SHOOTER (SPEAKER) The shooter mounts on a pivoting arm; the team tuned arm PID on 2024-02-17. The shooter wobbled in early tests, and the team built a new shooter iteration before the first regional. At Istanbul Regional the step inserts in the tube plug that held the shooter to the arm tore out; brackets replaced them.	**INTAKE** Touch-and-grab ground intake, a stated build goal. A 3847-style passive 3D-printed centering part was added on 2024-02-18 and kept. The intake motor changed from a 775-class motor to a NEO after motors burned out while notes stuck at the feeder. The steel intake plates bent repeatedly at Halic Regional; the team made its 8th or 9th plate change in the finals.
FEEDER / AMP / TRAP Feeder: the team tried 2 in 60A with 2.25 in 30A compliant wheels, then a NEO; notes still stuck. The fix was polyethylene film on the shooter bottom plate and a film-covered wedge; team 8159 recommended the film. Tennis racket grip on the front feeder roller helps feed the note. Amp: the hood design failed, so the robot shoots into the amp backward from the feeder with no hood.	**DRIVETRAIN** Swerve; the March 2024 update reports a battery dip to 7v when the swerve drive motors draw power. The team's 2025 blog says it is 'still using MK4i swerve drive'.
ENDGAME The climb design uses the arm. The team designed a chassis hook that latches on the chain to hold the robot. Its chain tensioner could not hold the climb loads, so the team did not climb at Istanbul Regional.	**AUTONOMOUS** At Istanbul Regional the robot had a 2-note auto and a 4-note auto that collects the three notes in front of the speaker. After the event the team had a 4-note auto with far shots and was developing a 5-note auto.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** 'Blackout' CAD (Onshape)

LESSONS FOR STUDENTS

- Check where the intake pushes the note before you change wheels or motors.
- Match plate material to impact loads: 3mm polycarbonate plates fixed both the bending and the loose centering pieces.
- Test the climb with full loads before an event; the chain tensioner failed in normal climbs, so the team did not climb at its first regional.
- Fasten the shooter to the arm with parts rated for the loads; tube-plug step inserts tore out at the first regional.
- Comm loss in five qualification matches stopped after the team changed the radio power module to a PoE injector cable. The team says it is not sure this was the cause.

Onshape CAD: 'Blackout' CAD (Onshape) · 'Blackout' CAD (Onshape) · Build blog and CAD release (CD) · 2025 build blog with offseason recap (CD)

Profile

2423 The KwarQs ConDUCKtor

USA-MA

Local EPA 24 (#731)

20-23

One pivot for shooter and climber; over-the-bumper intake; swerve

SHOOTER (SPEAKER) Pivoting shooter. The same pivot is used to climb. The team redesigned the shooter between DCMP and BattleCry at WPI.	**INTAKE** Over-the-bumper intake. The team went through several intake iterations during the competition season.
FEEDER / AMP / TRAP Unconfirmed	**DRIVETRAIN** Swerve, run with YAGSL; the team's second year using swerve drive. The release shares bolt-on MAXSwerve covers and 3D-printed TPU MAXSwerve wheels that the team developed in the 2023 offseason; the team states they served it well.
ENDGAME Climb with the pivoting shooter/climber.	**AUTONOMOUS** Autonomous modes built in PathPlanner. The team also tried PathPlanner pathfinding to drive to scoring positions in teleop.
VISION PhotonVision for AprilTag detection. The team also used the PhotonVision object-detection pipeline to automate picking up game pieces.	**PUBLIC CAD / DESIGN** Robot CAD (Onshape)

LESSONS FOR STUDENTS

- The configurable crush block for 2x1 tube labels each configuration by wall thickness and uses teardrop holes for 3D printing. The team later added a 1x1 version after forum feedback.
- Bolt-on MAXSwerve covers let the team clean and re-grease modules without taking them apart.
- The team moved from mostly custom code to YAGSL and command-based Java and states this got the swerve running early so it could focus on other subsystems.

Onshape CAD: Robot CAD (Onshape) · CAD and code release (CD) · Robot CAD (Onshape) · Code (GitHub) · Configurable crush block (Onshape) · 3D-printed bumper mounts (Onshape) · MAXSwerve screw-on covers (Onshape)

■ Study pick

5987 Galaxia in memory of David Zohar Andromeda

Israel

Local EPA 24 (#693)

30-26

Polycarbonate spring-hook Stage climber with a locking cable winch; flipped swerve

SHOOTER (SPEAKER) At the Israel District Championship the robot focused on the Speaker in a three-offense alliance. Before shipping to the Championship, the team spent most of its repair time on long-range shooting.	**INTAKE** In a District Championship playoff match the robot lost a belt on its intake and switched to defense.
FEEDER / AMP / TRAP At the District Championship the team describes delivering Notes to the Amp for its partners to score.	**DRIVETRAIN** Swerve with flipped modules. A Dec 2024 team post says the team used flipped swerve modules for the past two seasons.
ENDGAME Stage chain climb. Two polycarbonate strips, 8mm thick and 40mm wide, carry hooks at their ends. Cables run from the hooks to a winch gearbox that the team repurposed from its elevator. The cables hold the strips below Stage height; letting out cable lets the polycarbonate spring the hooks up. If the hooks rise too high, the team runs the cable into the Chain to pull them down until they catch. The robot winches in and locks the gearbox to stay up after T=0. Changed after Champs: for an October 2024 off-season event the team replaced the lock servo with a BAG motor.	**AUTONOMOUS** Two 4-Note routines, one on the near side and one on the far side of the Wing, shown in a team post during the district events. At that time the team said the far-side routine still needed work to finish under 15 seconds.
VISION Unconfirmed for the competition robot. In December 2023 the team's off-season vision setup used Orange Pi 5 coprocessors, Arducam OV2311 cameras and PhotonVision, plus a Limelight. A Dec 2024 team post says inefficient vision code caused loop overruns at an October 2024 off-season event.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A spring hook can be two polycarbonate strips that use the plastic's spring force to raise the hooks when the winch lets out cable.
- Design the hold after T=0 with care: the team says the gearbox lock was its only major mechanical problem at Champs.
- Reuse parts the team already trusts: the climb winch uses a gearbox the team repurposed from its elevator.
- Organize CAD for parallel work: before the season the team moved from SOLIDWORKS to Onshape and used one document per subsystem to limit load times and accidental changes.

Onshape CAD: Onshape CAD · Onshape CAD (Spectrum CAD Collection) · Build thread (CD) · Reveal thread (CD) · Reveal video · Off-season notes on the robot (CD, 2025 build thread)

■ Study pick

3737 Roto Raptors Synthwave

USA-NC

Local EPA 22 (#850)

25-18-2

Through-robot note path; under-bumper intake; sprung single-hook climber; no trap; 3D-printed swerve

SHOOTER (SPEAKER) The release post names a 'cannon' that rotates with torsion spring assist and raises the climber lift arm. The post does not say the cannon is the shooter.	**INTAKE** Under-bumper ground intake. The spring-loaded roller assembly rises for more ground clearance when the robot is not intaking. No mechanisms extend outside the chassis perimeter.
FEEDER / AMP / TRAP Through-robot note handling with dual belts to move the note. The team states the robot could do every game task except the trap.	**DRIVETRAIN** Compact swerve modules of the team's own design, printed in PETG on FDM printers. The release names NEO drive and NEO 550 steer motors as the base design, with SPARK MAX controllers; calculated free speed with a NEO is 15.41 ft/sec. Steering pairs the NEO 550 with UP 4:1 and 3:1 gearbox stages and turns on 1/4 in Delrin balls in printed PETG V-grooves. Each module has a low-cost absolute encoder; a 2024-01-29 update says to remove resistor R4 so the encoder works. Wheels are the team's 3D-printed TPU suspension treads on a printed PETG hub. The robot is under 14.5 in tall in starting configuration, which the team states permits maximum speed with no tipover. The team calls it its fastest robot. Changed for 2025 to NEO Vortex drive motors.
ENDGAME Single-hook chain climber. The hook is articulated and sprung to remove moment on the arm while hanging, and its profile resists slipping. The cannon raises the lift arm to reduce motor requirements. Compact needle roller bearings held by 3D prints guide the lift arm extension. Two servo-operated locks prevent a drop after the match. No trap.	**AUTONOMOUS** Unconfirmed
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Robot CAD (STEP, Google Drive)

LESSONS FOR STUDENTS

- An FDM 3D-printed swerve module can run a full season: the team reports no module failures, at about $200 per module without motors and controllers. The release includes a BOM, print settings, assembly steps and a speed calculator.
- Read the encoder notes before building: a 2024-01-29 update says the assembly instructions left out a link wire the encoder needs, and the team's fix is to remove resistor R4.
- Tune a printed tread to robot weight: the team sets TPU infill at 40% for a 90 lb robot and 50% near max weight, and reports a tread life of two competitions.
- Use one mechanism to help another: the cannon raises the climber lift arm, which reduces motor requirements.
- Design for service: two screws remove the bumpers, and clamshell 3D-printed cable management allows quick changes.

Drive folder / doc: Robot CAD (STEP, Google Drive) · Robot CAD (STEP, Google Drive) · Electrical board CAD (STEP, Google Drive) · CAD and code release thread (CD) · 3D-printed swerve release (CD) · 3D-printed suspension treads release (CD) · 3D-printed swerve files (Google Drive) · 3D-printed suspension tread files (Google Drive)

■ Study pick

6925 W.A.Robotics VENDETTA

USA-GA

Local EPA 22 (#878)

26-21

Fixed-angle 2-Kraken shooter; over-the-bumper intake with handoff; two hook climbers; MK4i swerve

SHOOTER (SPEAKER) Top and bottom shooter rollers, each powered by an independent Kraken. In January the team wrote that the shooter angle should be 60 degrees; it considered an adjustable shooter and chose not to build one. In February tests, 65% speed gave consistent speaker shots up to 14 inches away; higher speeds damaged notes and drew more power. Side shots and lofted long shots were less accurate.	**INTAKE** Over-the-bumper intake with custom plates cut on an X-Carve. The team states it picks up notes that are not centered. A limit switch is mounted on the intake with a custom bracket. The pivot motor uses an 80:1 MAXPlanetary gearbox, added in February. That gearbox was built wrong and was replaced in late February. In February the team started a second intake and reported it almost finished.
FEEDER / AMP / TRAP The intake hands the note to the shooter; a curved backplate added on 2024-01-31 holds the note for the handoff. Amp: the original plan was to score with the intake, which the team tested. The shooter also scores in the amp: 29 of 29 shots at 35% speed.	**DRIVETRAIN** Swerve with SDS MK4i modules. The team planned 8 Krakens and in February waited for the MK4i Kraken adapter kits with 16T drive pinions. Swerve code runs on Phoenix 6.
ENDGAME Two climbers with custom hooks. In early tests one climber ran faster than the other at the same speed; the team considered a churro to link them. On 2024-03-13 the team tested the hooks on the chain, including a climb on the side of the chain.	**AUTONOMOUS** The team tested 4-piece and 5-piece autonomous paths in January 2024, before the competition robot was finished.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Robot CAD (Onshape)

LESSONS FOR STUDENTS

- The team chose a fixed-angle shooter for a team of about 15 members and 6 veterans. Read the January post for the reasoning.
- The February shooter tests give a method a small team can copy: vary speed and position, record hit rate, and stop tests that fail early.
- Shooter speed is a trade: higher speed improved accuracy but damaged notes and drew more power, so the team set 65%.
- Two scoring paths for the amp, intake and shooter, give a backup if one subsystem breaks. The team tested both.
- One note shot 235 times showed visible wear. Keep new notes for competition tests.

Onshape CAD: Robot CAD (Onshape) · Robot CAD (Onshape) · Open Alliance build thread (CD) · Code (GitHub) · Reveal 'VENDETTA' (CD) · Reveal video

■ Study pick

4276 Surf City Vikings

USA-CA

Local EPA 19 (#1150)

15-9-1

Intake/shooter on a NEO-driven 4-bar, rebuilt mid-season; floor intake; climber on separate motors; MaxSwerve

SHOOTER (SPEAKER) Combined intake/shooter unit on a 4-bar arm. In January the team tested a left/right wheel shooter and reported it working. The team calls the competition shooter 'pretty much identical' to the wood prototype, with belts instead of chain. The 4-bar first used a modified REV Starter Bot drive tower at about 250:1 on one NEO. Changed after Ventura: doubled arms, 1/4 in mounting plates in place of MaxTube, two 4-bar motors.	**INTAKE** Floor intake with top/bottom wheels in the same unit as the shooter. A NEO 550 on 3:1 drives 2.25 in AndyMark compliant wheels. It stalled on floor pickup, so the team changed to 5:1 in week 6. The first intake was wood; the team replaced it with polycarbonate.
FEEDER / AMP / TRAP An E18-D80NK beam-break sensor detects the note in the intake/shooter. Reflection off polycarbonate and the aluminum brace blocked it; matte black grip tape on the brace fixed it. In a build-season test the team ran the upper roller in reverse to place notes in a mock amp.	**DRIVETRAIN** New MaxSwerve chassis. At Ventura the drive motor keys did not stay in place. At Orange County the team reports no swerve drive issues. During build season a bent pin on a through-bore encoder stopped one module's turning reading.
ENDGAME Two climber assemblies. At Ventura the climber was driven from the 4-bar, which the team calls disastrous. Changed for Orange County: the climber got its own motors, and the team reports it worked. A spring-loaded multihook add-on was in work in week 6; its use is unconfirmed.	**AUTONOMOUS** January plan: score in the speaker or amp, then leave.
VISION An auto shot alignment was added in the offseason, before Beach Blitz; its sensor is not named.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Stress-test the robot before the first event; the team lists skipping this as a cause of its Ventura failures.
- One NEO driving both sides of a 4-bar through a shaft was wobbly and failed; doubled arms and two motors held up at Orange County.
- Give each mechanism its own drive: the climber driven from the 4-bar failed at Ventura and worked on separate motors at Orange County.
- Check intake gearing under load: a NEO 550 at 3:1 stalled on floor pickup, and 5:1 worked.
- A beam-break sensor can fail on reflective surfaces; matte grip tape on the brace behind it fixed the reading.

Onshape CAD: Onshape CAD · Onshape CAD · Open Alliance build blog (CD) · Offseason build blog (CD)

■ Study pick

663 Power Sharks Hammerhead

USA-MA

Local EPA 17 (#1418)

18-23

Ri3D-based wide intake-shooter; sliding 8020 extending climbers on #35 chain

SHOOTER (SPEAKER) Combined intake-shooter, based on the Unqualified Quokkas Robot in 3 Days design. The team made it wider than the game piece to make intaking easier, and states this made accurate shooting harder.	**INTAKE** The intake is the wide intake-shooter. Its large span took hits at competition. The 6000-series churro tubing used to save weight yielded quickly; the team replaced it at the event with 7075 aluminum hex stock, which held up better. The team notes that some other Quokka-style robots put the intake inside the frame.
FEEDER / AMP / TRAP Unconfirmed	**DRIVETRAIN** The team used 8020 extrusion in many places instead of box tube, with through-bolted connections. The team states weight was not an issue for this design.
ENDGAME Sliding-contact extending climbers on 8020 slides, driven with #35 chain. The team states they support side-by-side buddy climbing on the chain. Through-bolts on the load path are needed because climb loads can make slide-nut connections slip.	**AUTONOMOUS** Unconfirmed
VISION No camera system is stated. Grapple Robotics laser sensors detect the game piece.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Size structural stock for impact, not only weight: 6000-series churro yielded on the wide intake, and 7075 hex stock replaced it at the event.
- Through-bolt the climb load path; the team states slide-nut connections can slip under climb loads.
- A wider intake-shooter trades shot accuracy for easier intaking, by the team's own account.
- A Robot in 3 Days design can be a season base: Hammerhead follows the Unqualified Quokkas Ri3D robot.

Onshape CAD: Onshape CAD · Onshape CAD · CAD release thread (CD)

Profile	1072 Harker Robotics USA-CA	Local EPA 16 (#1561) 11-19

Pivoting top-down shooter on a two-stage cascade elevator; over-the-bumper ground intake; centering indexer; MK4i swerve

SHOOTER (SPEAKER) Top-down shooter on a pivot mounted on the elevator. Early prototypes used Colson wheels as flywheels. A Falcon-powered prototype reached about 8 feet; adding Cat Tongue grip tape tripled the distance, but the tape flung loose.	**INTAKE** Over-the-bumper ground intake. The hand-run prototype used aluminum rollers with silicone tubing. The team found it was nearly impossible to shift notes left and right, so it needed a centralizing mechanism.
FEEDER / AMP / TRAP An indexer between the intake and the shooter centers the note and improves the handoff. The two-stage cascade elevator raises the pivoting shooter; the team chose this layout to score in the Trap.	**DRIVETRAIN** Swerve on SDS MK4i modules. The team switched from MK4 to MK4i, stating MK4i is safer for the Falcons. Modules bought before June used #8-32 mounting while later parts used #10-32, so mixing parts caused hole mismatches. The robot uses two CANivores, one per drivetrain half, because devices on longer stub lengths did not register.
ENDGAME The team's priority list includes a 2 Onstage climb and the Trap.	**AUTONOMOUS** In February 2024 the team planned a six note autonomous path.
VISION Limelight AprilTag pose estimation. The team loaded the field map into the Limelight pipeline and read robot poses from NetworkTables.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Let one scoring goal drive the architecture: the team chose an elevator-mounted pivoting shooter because it wanted to score in the Trap, and published its architecture comparison.
- Add a centering stage when the intake cannot center the game piece: the team found it nearly impossible to shift notes sideways in the prototype, so an indexer centers the note before the handoff.
- Check part revisions before mixing hardware: the team found MK4i parts ordered before June used #8-32 mounting while later parts used #10-32.
- Split a CAN FD bus when devices on long stubs do not register: the team used two CANivores, one per drivetrain half.

Onshape CAD: Onshape CAD · Onshape CAD · Build thread (CD) · Code (GitHub)

■ Study pick

1746 Team OTTO

USA-GA

Local EPA 16 (#1591)

30-17

Falcon shooter fed by a primer; intake rebuilt on a Falcon; CTRE swerve

SHOOTER (SPEAKER) Shooter rollers on Falcon motors. The team states SysID worked for the shooter: they entered the Falcon constants and it worked at once.	**INTAKE** Intake v1 was ineffective; drivers often had to push notes against a wall to intake. It burned 3 NEO motors and their SPARK MAX controllers with current limits applied. A Falcon replaced them and worked.
FEEDER / AMP / TRAP A primer feeds the shooter. It ran on a 775 motor, which did not have enough torque or speed to shoot the note consistently. Primer feed was inconsistent and sometimes hit the rollers, so the team could not reliably run the intake sequence with the shooter rollers running. At the District Championship the team stored notes in the intake while ferrying as a workaround.	**DRIVETRAIN** Swerve, run with CTRE swerve code that the team states integrated without problems.
ENDGAME Unconfirmed	**AUTONOMOUS** Paths built with PathPlanner and Choreo; the team states both integrated well.
VISION PhotonVision, used in simulation during development.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Size motors to the load: a 775 on the primer could not shoot consistently, and the NEO intake burned out three times before a Falcon replaced it.
- Test the intake early against open-floor notes: v1 needed notes pushed against a wall to pick them up.
- Name commands by how they end: the team found commands with similar names that ended at once, ran forever, or ran until a condition.
- SysID characterization of the shooter worked on the first try with the Falcon constants, per the team.
- Quick merges at events led to redundant code and bloated subsystems, per the team's reflection.

Onshape CAD: Onshape CAD · Onshape CAD · Code (GitHub) · Code and CAD release with reflection (CD) · DCMP recap video

Profile	9047 SESI SC TECHMAKER Seu Wilson Brazil	Local EPA 15 (#1620) 7-3-1

Fixed under-bumper intake-shooter; elevator that climbs and carries an Everybot-style amp claw; MK4i L3 swerve

SHOOTER (SPEAKER) Fixed intake-shooter inspired by the RI3D Quokkas robot. The design goal was to score in the Speaker from the Stage, and the team reports consistent Speaker shots with the prototype. In early February the team reported a weakness in the shooter and said it was making it stronger.	**INTAKE** Wide 'touch it, own it' ground intake below the bumper, combined with the fixed shooter. The team planned a sensor or switch to detect when a note is inside.
FEEDER / AMP / TRAP A claw on the elevator, based on the Everybot 118 claw, picks up from the Source and delivers to the Amp.	**DRIVETRAIN** Swerve on MK4i L3 modules with NEOs, driven in code with YAGSL. Drivetrain 28 in, initial robot height 26 in, low enough to pass under the chain. The design target was a weight below 40 kg.
ENDGAME The elevator is also the climber, with a climb height of 34.6 in. It uses the Compact Elevator Bearing Kit, HTD 5m belts, ABS printed pulleys and a MAXPlanetary gearbox. In early climb tests the belt shifted in its clip, and the team planned a printed HTD 5M clamping plate. The must-have list included hanging on the chain with a partner.	**AUTONOMOUS** Build-season goal of 3 notes, 4 if possible, plus a test route to collect 2 notes from the center of the field.
VISION Build-season plan: use AprilTags to navigate to field points, mainly the Source, with a Raspberry Pi co-processor. The team also planned note and robot detection to avoid collisions in auto.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Write the must-have list before CAD: MK4i L3 under 40 kg, floor pickup, a wide under-bumper intake, Speaker shots from the Stage, Amp scoring, and a chain hang with a partner.
- One elevator, two jobs: it climbs and carries the amp claw, and the robot stays low enough to pass under the chain.
- Belt-driven elevators need a secure belt clamp under climb load: the belt shifted in its clip in early climb tests.

Onshape CAD: Onshape CAD · Open Alliance build thread (CD) · Onshape CAD · Code (GitHub)

■ Study pick

3596 Rocket Robotics Kerchunk

USA-WI

Local EPA 11 (#2329)
3-7

Kitbot launcher; under-bumper ground intake; AndyMark Climber in a Box; 27x27 in swerve chassis

SHOOTER (SPEAKER) Based on the Kitbot launcher. The build thread states NEO motors on the Kitbot shooter during testing; motors on the final robot unconfirmed. In Kitbot testing the team saw a left hook and filmed it in slow motion. The note decompressed while still touching the right shooter rail, and the team planned to change where that rail ends. In the off-season the team added a second set of shooter wheels, which it says improved performance.	**INTAKE** Under-the-bumper ground intake.
FEEDER / AMP / TRAP Unconfirmed	**DRIVETRAIN** 27x27 in swerve chassis. The team states the 27x27 size fits through its doors with standard bumpers.
ENDGAME The students chose two 2-stage AndyMark Climbers in a Box, with modified hooks.	**AUTONOMOUS** The team used YAGSL and PathPlanner for the first time.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Start from a known base: the team kept the Kitbot launcher and changed the chassis, intake and climber around it.
- Film the failure: slow-motion video showed the note decompressing against the right rail, which the team took as the likely cause of the Kitbot left hook.
- COTS climbers can carry an endgame: two 2-stage AndyMark Climbers in a Box, with modified hooks, gave the climb.
- Iterate after the season: an off-season second set of shooter wheels improved performance, per the team.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Open Alliance build thread (CD)

Sources: the Spectrum 3847 CAD Collection, team CAD, binder and code releases, Chief Delphi release and reveal threads, and match data from The Blue Alliance. EPA is Local EPA (approx), computed independently. It is not official Statbotics data. "Unconfirmed" means the public material does not state the detail.